

CO₂ sequestration and mineralization in basalts: Insights from a deep learning-based surrogate model

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ABSTRACT

Basalt formations emerge as promising Carbon Dioxide (CO₂) sinks facilitating rapid mineralization that transforms CO₂ into a solid state and mitigates leakage risks. However, mineralization processes at an industrial scale present significant uncertainty, and traditional numerical simulations are time-consuming and resource-intensive. To address this, we propose a surrogate model to accelerate the prediction of carbonate precipitation by integrating deep learning proxies. Our model incorporates seven initial minerals, four precipitation minerals, and the critical factor of porosity changes. It achieves computational efficiency that is 20,000 times faster than traditional numerical models while maintaining high accuracy. Using a combined approach of surrogate and numerical models, we conducted a parameter analysis across multiple injection scenarios. We found that the porosity of the reservoir near the injection well undergoes a self-enhancing to self-sealing process, which leads to a decline in injectivity during the later stages of injection. However, placing the injection well at the top of the reservoir can mitigate this issue, reducing the decline in injectivity and minimizing pressure buildup. From the perspective of storage efficiency, positioning the injection well at the bottom of the reservoir and lowering the injection rate enhances the overall efficiency of CO₂ sequestration. Notably, magnesite exhibited the highest accumulation around the injection well, potentially having the most significant long-term impact on reservoir stability. This advancement highlights the promise of deep learning techniques in accelerating and enhancing the assessment of CO₂ mineralization processes in Carbon Capture and Storage (CCS) projects, offering a computationally feasible solution for forecasting complex geochemical interactions.

1. Introduction

In recent years, the concentration of Carbon Dioxide (CO₂) in the global atmosphere is higher than at any point in human history (Blunden and Arndt, 2019). Given these pressing issues, addressing CO₂ emissions has become an urgent priority. Among various mitigation strategies, Carbon Capture and Storage (CCS) plays a critical role in achieving the goals of the Paris Agreement. The CCS system comprises several stages, including capture, separation, transportation, storage, and monitoring (Snæbjörnsdóttir et al., 2020). Effective storage solutions ensure that captured CO₂ is securely sequestered, preventing its release back into the atmosphere and thereby mitigating the adverse impacts of climate change.

Sequestered CO₂ primarily undergoes structural trapping followed by residual trapping, solubility trapping, and mineral trapping,

converting from a gaseous state to solid carbonates that are permanently stored in the subsurface (Snæbjörnsdóttir et al., 2020). During these stages, the primary concern is the safety of the storage specifically, whether CO₂ might leak, causing secondary pollution. Among all trapping mechanisms for sequestered CO₂, mineral trapping through carbonation is regarded as the safest (Bachu, 2008). The choice of storage reservoir can significantly influence the outcome. Sandstone, due to its lack of reactive minerals, has limited capacity for mineral trapping in the short term, reducing its potential for permanent CO₂ storage. In contrast, basalt is rich in calcium, magnesium, and iron-bearing minerals, which can react with dissolved CO₂ to form stable carbonates, providing great potential for secure long-term CO₂ storage. Recently, several engineering projects have demonstrated successful CO₂ mineralization in basalt formations. The CarbFix project in Iceland showed that over 95 % of injected CO₂ was converted to carbonate

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minerals in less than 2 years, significantly faster than previously expected (Matter et al., 2016). Similarly, a field demonstration in eastern Washington injected 1000 metric tons of CO₂ into basalt, converting it to carbonates within 2 years, validating the rapid CO₂ mineralization observed in lab tests (McGrail et al., 2017). The Jizan region of south-west Saudi Arabia has the potential to dispose of substantial quantities of CO₂ through the subsurface mineralization of local basaltic rocks, with ongoing injection efforts further supporting this potential (Oelkers et al., 2022).

Given these promising results, accurately and quickly predicting CO₂ sequestration in basalt formations is crucial for future projects. Numerical simulations are essential tools in this evaluation, as they help to understand and predict the behavior of CO₂ in subsurface environments. However, these simulations involve not only multiphase flow transport but also complex mineral chemical reactions, making them very time-consuming and computationally expensive (Doughty, 2010). The numerical models adhere to rigorous governing Partial Differential Equations (PDEs), which represent high-fidelity physical processes. Consequently, when dealing with site selection, injection optimization, and uncertainty analysis, a large number of forward numerical simulations need to be conducted (Chen et al., 2020; Kitanidis, 2015).

To address these challenges and avoid the complexities of solving detailed physical processes, deep-learning based surrogate models have been proposed as beneficial alternatives. These models provide a simplified yet accurate representation of the underlying physical processes and can significantly reduce the computational burden associated with numerical simulations. This reduction makes it feasible to perform the extensive evaluations required for CO₂ sequestration projects. Numerous studies have applied image-based approaches and snapshots of simulation data over a spatially discretized input domain for surrogate modelling of subsurface flow and transport problems. Among these approaches, Convolutional Neural Networks (CNNs) have achieved significant success due to their ability to extract multi-scale features through multiple convolutional layers (Tariq et al., 2023; Wang and Lin, 2020; Yan et al., 2024). In multiphase flow transport models in porous media, changes in velocity and pressure fields are primarily determined by the local neighbourhood states. CNNs perform local operations through convolutional kernels, with weights shared across the entire input space. The solution can be represented as a mapping from input conditions (such as initial or boundary conditions) to the solution itself. This is akin to image-to-image regression tasks, where CNNs approximate the complex nonlinear relationship between input feature maps (such as permeability fields) and output state variables, thereby improving prediction accuracy and efficiency. Additionally, the network structure based on Fourier Neural Operators (FNO) has been extensively studied (Jiang et al., 2024; Wen et al., 2022). One of the notable advantages of FNO is its resolution-invariant property, which allows it to be trained and applied to input-output data of varying resolutions without needing to retrain the model. This characteristic underscores its advantages in multi-scale modelling of multiphase flow transport. Through Fourier transformation and inverse Fourier transformation, FNO can effectively capture and represent the complex features of the flow process, filtering out high-frequency noise while retaining essential low-frequency information. Furthermore, Long Short-Term Memory (LSTM) networks have been used for predicting multiphase flow (Feng et al., 2024; Xie et al., 2023). LSTM models can predict the complex long-term evolution of state variables using only a portion of the training data. They demonstrate the ability to extrapolate to future time steps that are unseen during the training process. Through their multi-layer structure and memory units, LSTMs are capable of capturing features and patterns at different time scales, thereby enhancing the understanding and predictive capability of the entire system's behavior.

However, most deep learning surrogate models primarily focus on multiphase flow, with limited consideration of chemical reactions. The inclusion of chemical reactions is more challenging due to the higher degree of coupling in the PDEs. Existing studies are, such as those by

Tariq et al. (2023), have only considered a limited number of reactions in saline aquifers and included simulations of only a few minerals. This highlights a significant gap in the current research and underscores the need for models that can effectively handle the complexities of both multiphase flow and chemical reactions in basalt CO₂ sequestration. Additionally, we have accounted for the critical impact of porosity changes during the sequestration process, which directly influences permeability. This, in turn, alters the stress field and significantly affects the overall stability of the formation (Chen et al., 2024a; Xin et al., 2024).

In this study, a deep learning-based surrogate model is developed to address the challenges of CO₂ sequestration in basalt formations. By incorporating seven initial minerals and four secondary products, we simulate a total of 11 minerals, offering a comprehensive and realistic representation of the complex geochemical interactions involved. The impact of mineralization reactions on porosity is also considered, as changes in porosity influence both injection efficiency and long-term storage safety. The deep learning model significantly accelerates the prediction process, being 20,000 times faster than conventional numerical simulators. Through a combined approach using surrogate and numerical models, we analyzed multiple injection scenarios. Our results indicate that porosity near the injection well undergoes a self-enhancing phase, where porosity increases, followed by a self-sealing phase, where porosity decreases. The placement of the injection well plays a crucial role in these dynamics. Injecting CO₂ at the bottom of the reservoir promotes greater dissolution and mineralization, improving the overall sequestration efficiency. However, this strategy also raises the risk of pressure buildup and self-sealing. On the other hand, injecting CO₂ at the top of the reservoir helps mitigate pressure-related risks and reduces the likelihood of self-sealing, making it a safer option for long-term storage. The spatial distribution of minerals revealed that magnesite, a key precipitate in the sequestration process, forms closer to the wellbore and exhibits the largest amount of precipitation. As a result, it is likely to have the greatest impact on the reservoir's stability and long-term integrity. Our work utilizes the computationally efficient deep learning approach for sensitivity analysis requiring thousands of iterations. Deep learning and simulators are complementary: deep learning provides rapid initial insights, while simulators remain indispensable for final validation.

This manuscript is structured as follows. Section 2 primarily introduces the establishment of the numerical model, including the governing equations for multiphase flow and chemical reactions, the selection of model parameters, benchmark model and field-scale model. Section 3 provides a detailed description of the neural network architecture and training strategy. Section 4 presents the results, showcasing predictions of mineral precipitation and porosity changes, as well as an evaluation of the model's accuracy. In Section 5, a combined approach using the surrogate model and numerical simulations is employed to perform a parametric analysis across various injection scenarios. Finally, Section 6 summarizes the study.

2. Numerical model

2.1. Problem setup

We developed a field-scale model as shown in Fig. 1a, to analyze the spatial and temporal distribution of minerals and their impact on geological formations (Liu et al., 2019).

The numerical model for this site is based on the Columbia River Basalt Group, where supercritical CO₂ is injected into a composite injection zone (775–875 m). The simulation duration is set to one year, which is sufficient to capture significant mineralization due to the relatively rapid reaction between basalt and CO₂. The Columbia River Basalt Group is conceptualized as a cylindrical domain, as shown in Fig. 1b, with a thickness of 100 m and a radius of 1200 m. To accurately capture fluid flow dynamics, 100 logarithmic grid cells were used in the

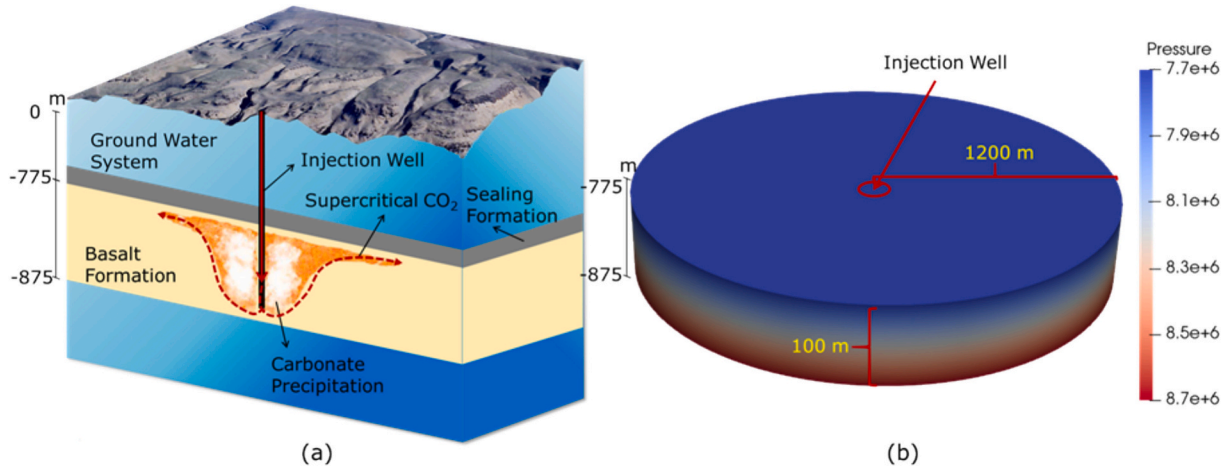


Fig. 1. (a) Cross-section of the conceptual model: A schematic illustration depicting the injection of supercritical CO₂ into the underground basalt formation at depths between -775 m and -875 m. (b) Overview of the numerical model: The model conceptualizes the domain as a cylindrical region with a height of 100 m and a radius of 1200 m.

radial direction and 30 uniformly spaced grid cells were used in the vertical direction, resulting in a total of 3000 grid cells. A grid independence test was conducted and the details are provided in Appendix A. The top and bottom boundaries are set as no-flux boundaries to simulate impermeable layers, while the outer radial boundary is set at constant pressure and temperature, located sufficiently far from the injection well to mimic an infinite-acting reservoir.

Within the composite injection zone, initial conditions, consistent with field measurements from the Wallula Basalt Carbon Storage Pilot Project (Jayne et al., 2019; McGrail et al., 2009), include a top temperature of 35 °C with an initial pressure of 7.7 MPa, and a bottom temperature of 38 °C with an initial pressure of 8.7 MPa. The hydrogeologic and thermal properties of the basalt formation are derived from data collected during the Wallula site characterization (McGrail et al., 2009). Parameters for relative permeability and capillary pressure are described by the Van Genuchten model (Pollyea, 2016), and are summarized in Table 1.

2.2. Governing equations

The geological CO₂ sequestration model we developed is coupled with subsurface solute transport, multiphase fluid flow, and chemical reactions to simulate the multiphase flow and geochemical interactions. We performed the simulations using TOUGHREACT, version 1.2. Flow and transport processes are described by mass conservation (Pruess, 1991; Xu et al., 2008):

$$\frac{\partial M}{\partial t} = -\nabla \cdot F + q \quad (1)$$

where M [kg m⁻³] is the mass accumulation, F [kg m⁻² s⁻¹] is the mass flux and q [kg m⁻³ s⁻¹] is the source/sink term. We implemented a porosity-permeability coupling using a power law (Steeff and Lasaga, 1994; Xu et al., 2008) to reflect permeability reduction due to mineral precipitation. This relationship is expressed as:

$$k = k_0 \times \left(\frac{\phi}{\phi_0} \right)^\alpha \quad (2)$$

where k [m²] is the updated permeability, k_0 [m²] is the initial permeability, ϕ is the current porosity, ϕ_0 is the initial porosity, and α is an empirical exponent. The power law is a widely accepted permeability–porosity relationship, supported by studies on porous media (Civan, 2001; Luquot and Gouze, 2009; Yang and Dunham, 2021). This relationship allows permeability to dynamically evolve with porosity changes during mineralization and dissolution processes. We chose $\alpha = 3$ following Wu et al. (2021), who showed that results are relatively insensitive to the choice of α , while omitting this relationship may lead to significant inaccuracies. Our choice balances physical realism with computational efficiency.

To effectively represent a geochemical system, it is useful to select a subset of aqueous species as basis species. The remaining species, known as secondary species, include aqueous complexes, precipitated minerals, and gaseous species (Reed, 1982; Steeff and Lasaga, 1994; Yeh and Tripathi, 1991). And the kinetic rate r_n of mineral dissolution or precipitation can be described by Eq. (3) as given by Lasaga et al. (1994):

$$r_n = f(c_1, c_2, \dots, c_i, \dots, c_N) = \pm K_n A_n [1 - \Omega_n^\theta] \quad (3)$$

where positive values of r_n [mol s⁻¹] indicate dissolution, and negative values denote precipitation. A_n [m² kg⁻¹ H₂O] indicates the specific reactive surface area per kilogram of H₂O, and Ω_n^θ is the kinetic mineral saturation ratio. The parameters θ and n are typically assumed to be equal to one. And c_i [mol kg⁻¹] is molal concentration of the i^{th} basis species. The temperature dependent term K_n [mol m⁻² s⁻¹] represents the rate constant and can be expressed as:

$$K_n = K_{25}^{nu} \exp \left[\frac{-E_a^{nu}}{R} \left(\frac{1}{T} - \frac{1}{298.15} \right) \right] + K_{25}^H \exp \left[\frac{-E_a^H}{R} \left(\frac{1}{T} - \frac{1}{298.15} \right) \right] a_H^{n_H} + K_{25}^{OH} \left[-\frac{E_a^{OH}}{R} \left(\frac{1}{T} - \frac{1}{298.15} \right) \right] a_{OH}^{n_{OH}} \quad (4)$$

where the superscripts or subscripts nu , H , OH denote neutral, acid, and base mechanisms, respectively. In this equation, R [J mol⁻¹ K] is the gas constant, T [K] is the absolute temperature, E_a [J mol⁻¹] represents the activation energy, and K_{25} [mol m⁻² s⁻¹] is the reaction rate constant at 25 °C. At each time step, the total reaction rate for each mineral is determined by calculating the reaction rate constants, the reactive surface area, and the kinetic mineral saturation state. These factors are

Table 1

Van Genuchten model parameters.

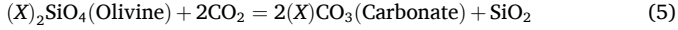
Parameter	Value	Unit
Residual Water Saturation	0.30	
Residual Gas Saturation	0.25	
Phase Interference Parameter	0.550	
Saturated Liquid Saturation	1.0	
Maximum Capillary Pressure	1×10^7	Pa

combined to evaluate the overall rate at which mineral dissolution or precipitation occurs within the geochemical system.

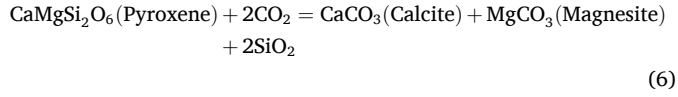
2.3. Selection of mineral parameters

In this study, we specifically used an unaltered flood basalt with relatively high olivine content to represent a reactive end-member. Olivine was chosen as the primary reactive phase due to its significantly higher reactivity compared to other basaltic minerals, such as pyroxene and plagioclase (Banfield et al., 1991; Craig and Loughnan, 1964; Dou et al., 2024; Gudbrandsson et al., 2011). This makes olivine-driven reactions dominant during the early stages of CO₂ injection, especially within the one-year timeframe considered in our simulation. Additionally, flood basalt is widespread on the Earth's surface (Rudnick and White, 2018). In the United States, the Columbia River Basalt Group (CRBG) spans more than 164,000 km² across Washington, Oregon, and Idaho (McGrail et al., 2006).

The primary mineralogy and specific surface area (SSA) values of the flood basalt were calibrated based on the work of Xiong et al. (2017). Detailed primary mineral compositions used in the simulation are provided in Table 2. For the secondary minerals settings, since olivine reacts the fastest, the secondary minerals were selected mainly as the byproducts of olivine's carbonation reaction (Kelemen and Matter, 2008):



Here, X represents either magnesium (Mg) or iron (Fe), indicating that the expected products are magnesite (MgCO₃) and siderite (FeCO₃), along with silicon dioxide (SiO₂). Additionally, the carbonation of Ca-rich pyroxene follows the reaction path:



This indicates that calcite (CaCO₃) may precipitate in addition to magnesite and siderite (Kelemen and Matter, 2008). Their selection was supported by experimental observations using Raman spectroscopy by Xiong et al. (2017), which confirmed their formation during CO₂-basalt reactions. Literature-reported SSAs were selected for all secondary minerals. The details of the secondary minerals are also listed in Table 2.

The reaction rate and activation energy (Ea) for each mineral were obtained from available literature sources (Table 3).

We also benchmarked our algorithm and mineral chemical parameters with the results presented in the appendix A.

Table 2
Primary and secondary mineral volume fraction and specific surface area.

Primary mineral	Initial volume fraction (%)	Specific surface area (cm ² /g)
Diopside (CaMgSi ₂ O ₆)	14 ^a	34 ^a
Hedenbergite (CaFeSi ₂ O ₆)	8 ^a	34 ^a
Albite (NaAlSi ₃ O ₈)	13 ^a	225 ^a
Anorthite (Ca(Al ₂ Si ₂ O ₈))	19 ^a	225 ^a
Forsterite (Mg ₂ SiO ₄)	8 ^a	1800 ^a
Fayalite (Fe ₂ SiO ₄)	5 ^a	1800 ^a
K-feldspar (KAlSi ₃ O ₈)	33 ^a	190 ^a
Secondary mineral		
Calcite (CaCO ₃)	0	370 ^b
Magnesite (MgCO ₃)	0	662 ^b
Siderite (FeCO ₃)	0	1050 ^c
Amorphous silica (SiO ₂)	0	225 ^d

^a (Xiong et al., 2017).

^b (Pokrovsky et al., 2005).

^c (Pokrovsky and Schott, 2002).

^d (Tester et al., 1994).

3. Deep learning based surrogate model

3.1. Sampling scheme

To train the deep learning model to predict the state variables of the formation under given operating conditions and site parameters, the generated dataset should include various operational and parametric information of the formation. By sampling the parameter space, the model can learn a generalized mapping relationship. The parameter space for this study's datasets includes randomly generated permeability and porosity fields, as well as random injection scenarios including location and rate.

The realizations of random fields are generated using the open-source Python library GStools (Müller et al., 2022). Each geologic realization is implemented with a distinct injection scheme. The injection well spans 10 grid cells, equivalent to 33 m in length, with a random perforation location. The total injection rates are sampled from a Latin hypercube distribution, ranging between 3 and 5 kg of CO₂ per second.

In total, we generated 1155 distinct cases, capturing a broad spectrum of potential operational conditions and reservoir characteristics. These cases are split into training and testing datasets in a 10:1 ratio. The training dataset consists of 1050 cases, while the testing dataset includes 105 cases. This sampling approach allows for robust model training and testing, ensuring that the surrogate models developed are capable of accurately predicting outcomes under a wide range of conditions.

3.2. U-Net with attention architecture

Drawing from the Diffusion U-Net (Rombach et al., 2022), we design an enhanced U-Net with attention structure, as shown in Fig. 2. The backbone of our deep learning structure is U-net. It consists of a spatial down-sampling process followed by a spatial up-sampling process, with additional skip connections between the down-sampling activations and the corresponding up-sampling layers. The down-sampling process captures broader context and structural features, while the up-sampling process refines this information and integrates local details through skip connections. These connections preserve high-resolution features, enabling U-net to achieve accurate image segmentation and reconstruction.

3.3. Self-attention mechanism

The self-attention mechanism, initially proposed in the Transformer architecture by Vaswani et al. (2017), allows the network to weigh the importance of different input elements. Following the Vision Transformer (ViT) approach (Dosovitskiy et al., 2020), we apply this mechanism to embed continuous functions or signals instead of word sequences.

To handle 2D data, we first reshape the data $x \in \mathbb{R}^{H \times W \times C}$ into a sequence of flattened 2D patches $x_p \in \mathbb{R}^{N \times (P^2 \cdot C)}$, where (H, W) is the resolution of the original image, C is the number of channels, (P, P) is the resolution of each image patch, and $N = \frac{HW}{P^2}$ is the resulting number of patches, which also serves as the effective input sequence length for the Transformer.

In practice, we convert the U-Net encoder output into a sequence of patches with sequence length N and embedding size D , represented as a matrix (Fig. 3a). These patches are then linearly transformed to obtain the key (K), query (Q), and value (V) matrices. Based on the number of heads, we split K , Q , and V into h groups of matrices. Each group of matrices has a length of d_k :

$$d_k = \frac{D}{h} \quad (7)$$

We compute the attention function on a set of queries simultaneously, packing them into a matrix Q . The keys and values are also

Table 3
Mineral kinetic parameters.

Mineral	Acid mechanism			Neutral mechanism		Basic mechanism		
	log k_{25} (mol/m ² /s)	Ea (kJ/mol)	n^{H^+}	log k_{25} (mol/m ² /s)	Ea (kJ/mol)	log k_{25} (mol/m ² /s)	Ea (kJ/mol)	n^{OH^-}
Diopside ^b	−6.36	96.1	0.71	−11.11	40.6			
Hedenbergite ^a	−6.36	96.1	0.71	−11.11	40.6			
Albite ^c	−8.86	67.7	0.5	−12	66.5			
Anorthite ^c	−8.86	67.7	0.5	−12	66.5			
Forsterite ^b	−6.85	67.1	0.47	−10.64	79.0			
Fayalite ^d	−6.85	94.3	0.47	−12.8	94.3			
K-feldspar	−9.45 ^e	51.9 ^b	0.5 ^b	−12.41 ^b	38.0 ^b	−21.2 ^b	94.1 ^b	−0.823 ^b
Calcite ^b	−0.3	14.4	1.0	−5.81	23.5			
Magnesite ^b	−6.38	14.4	1.0	−9.34	23.5			
Siderite ^f	−3.74	56.0	0.9	−8.90	62.7			
Amorphous silica (SiO ₂) ^g				−9.70	79.0			

^a The kinetics of diopside are assigned to hedenbergite (Palandri and Kharaka, 2004; Xiong et al., 2017).

^b (Palandri and Kharaka, 2004).

^c Albite and antigorite kinetics were set to those of labradorite, consistent with the plagioclase composition of both samples (Knauss et al., 2005; Xiong et al., 2017).

^d (Xiong et al., 2017).

^e (Knauss et al., 2005).

^f (Golubev et al., 2009).

^g (Carroll et al., 2013).

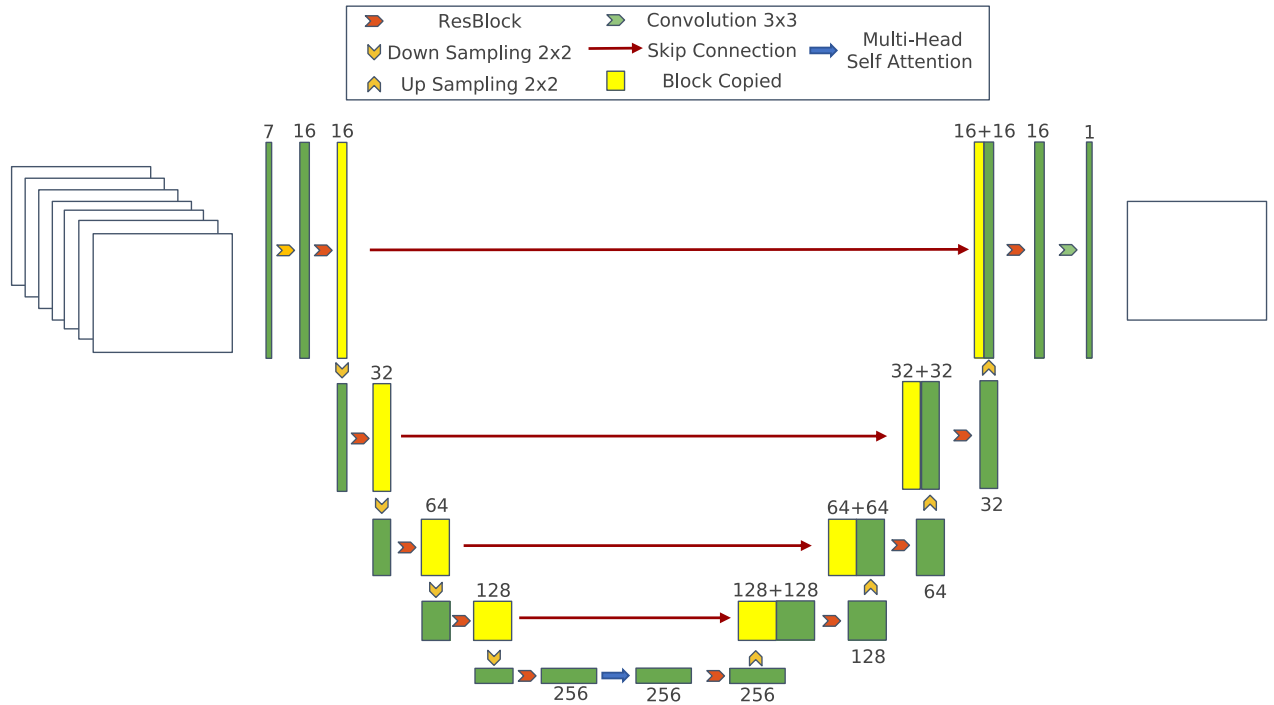


Fig. 2. U-Net with attention structure. The input has 7 channels, and the output 1 channel. The numbers on each block represent the number of channels at different stages.

packed into matrices K and V . The output matrix is computed as follows:

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (8)$$

The detailed processing sequence of the self-attention mechanism can be found in Fig. 3b.

Inspired by the design of TransUNet (Chen et al., 2021), we have incorporated the self-attention mechanism into the bottom layer of our network to capture spatial dependencies (Fig. 2). As the data has already been down-sampled, adding attention here avoids excessive parameter growth and overfitting.

3.4. Training strategy

In our model, the image-to-image prediction problem uses [permeability, porosity, grid volume, injection scenario, time, z-coordinate, x-coordinate] as inputs to predict [siderite volume, magnesite volume, calcite volume, and porosity increment]. All the input features are normalized into 0 and 1 by min-max scaling. The model processes seven input parameters structured as multi-channel 2D images ($7 \times 30 \times 100$), where each channel represents a distinct physical parameter mapped across the reservoir's vertical cross-section. The input of the deep learning surrogate model is illustrated in Fig. 4a-g which provide comprehensive physical and spatial context: permeability, porosity, grid volume, and coordinate maps describe the geological and geometric framework; the injection scenario channel encodes boundary conditions

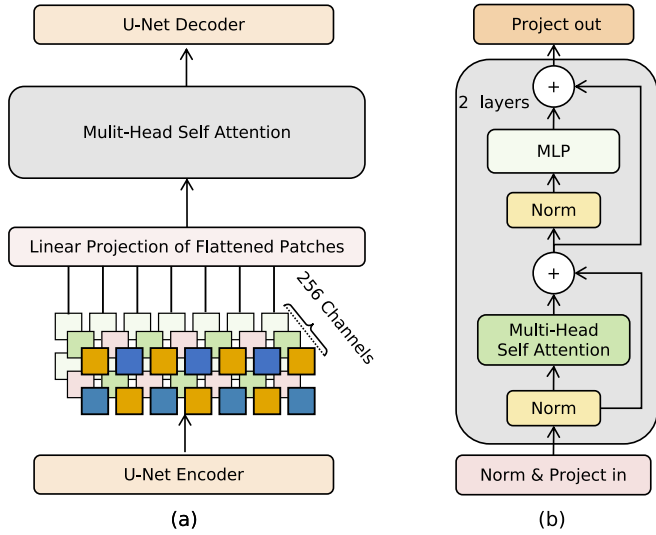


Fig. 3. Diagram of the self-attention algorithm. (a) The output from the U-Net encoder, with dimensions $h \times w = 2 \times 7$ and 256 channels, is processed by the multi-head self-attention mechanism before being passed into the decoder. (b) Detailed explanation of the multi-head self-attention mechanism.

with non-zero values only at the injection perforation (all other pixels are zero); and the time channel contains a constant value representing the specific timestep, enabling direct prediction at arbitrary times without sequential modelling. We trained four separate deep learning models, each predicting one of the target outputs: siderite volume, magnesite volume, calcite volume, and porosity increment. Each model outputs a $1 \times 30 \times 100$ image, capturing the spatial distribution of a specific property. This architecture allows the surrogate models to effectively learn the coupled interactions of fluid flow, reaction kinetics, and spatial heterogeneity over time while maintaining high computational efficiency.

We compare not only the differences between the predictions and the ground truth (simulations) but also the differences in their derivatives (Czarnecki et al., 2017; Feng et al., 2024). Consequently, the loss function we adopt is as follows:

$$L(\theta) = \frac{1}{n_s} \sum_{i=1}^{n_s} \|y_i - \hat{y}_i\|_p + \lambda \frac{1}{n_s} \sum_{i=1}^{n_s} \times \sum_{j=1}^k \|D^j y_i - D^j \hat{y}_i\|_p \quad (9)$$

where n_s represents the number of training samples, k denotes the

highest order of derivatives, y_i is the ground truth state variable, $\hat{y}_i$ is the prediction made by the surrogate model, D^j denotes the j order derivative with respect to spatial coordinates, and $\|\bullet\|_p$ represents the p -norm. The coefficient λ represents the weight of the spatial derivative term. In subsequent experiments, we set λ to 0.1 and k to 1, considering only the first-order derivatives.

To efficiently compute the derivatives, we utilize non-trainable 3×3 convolutional filters, which apply central difference discrete numerical approximations based on the finite difference method (Feng et al., 2024). The goal of training is to find θ^* by minimizing $L(\theta)$, shown as (10).

$$\theta^* = \underset{\theta}{\operatorname{argmin}} L(\theta) \quad (10)$$

The optimal deep learning models were identified by adjusting the hyperparameters of the convolutional neural networks, as summarized in Table 4. In our training process, we utilized the Adam optimizer to update the model parameters and incorporated a learning rate scheduler. The proposed neural network was implemented with the open-source deep learning framework PyTorch (Paszke et al., 2019).

4. Results

4.1. Model performance evaluating metrics

To better evaluate the performance of the model, we primarily use two dimensionless metrics: the coefficient of determination (R^2) and the structural similarity index measure (SSIM). These metrics were chosen to comprehensively evaluate both the numerical accuracy and the spatial fidelity of the predictions. R^2 is widely used in regression tasks and provides a quantitative measure of how well the predicted values approximate the ground truth, making it suitable for assessing mineral content. SSIM, on the other hand, evaluates perceptual similarity

Table 4
Summary table of hyperparameters.

Hyperparameters	Value
Batch size	32
Learning rate	5×10^{-4}
Weight decay	1×10^{-4}
Epochs	100
Patience	20
Reduction factor	0.8
Threshold	0.005
Minimum learning rate	1×10^{-5}

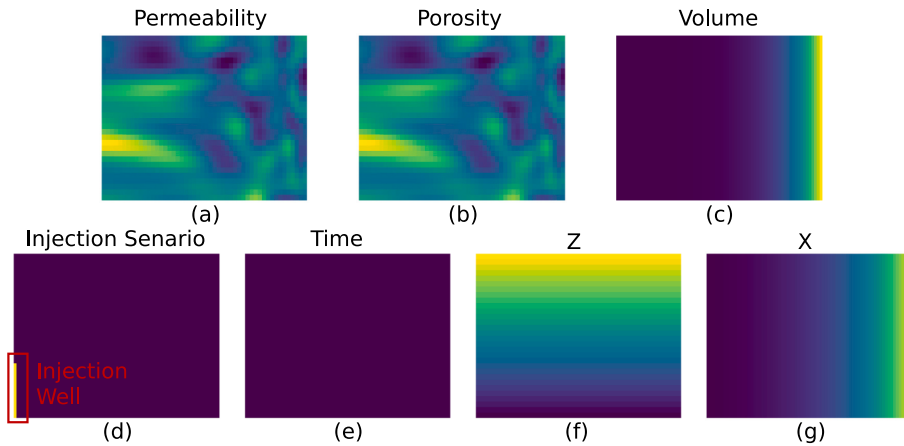


Fig. 4. Seven input channels of the deep learning surrogate model. (a) Permeability of each grid cell. (b) Porosity of each grid cell. (c) Volume of each grid cell. (d) Injection scenario: only injection wells have non-zero values, representing the injection rate, while other regions are set to zero. (e) Time parameter of each grid cell, representing the current time step. (f) Z-coordinate of each grid cell. (g) X-coordinate of each grid cells.

between predicted and actual spatial distributions, capturing structural information. This makes it particularly appropriate for assessing the accuracy of spatial-temporal mineralization maps. The definitions of each metric are provided below:

$$R^2 = 1 - \frac{\sum_{i=1}^{n_e} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n_e} (y_i - \bar{y})^2} \quad (11)$$

where n_e is the number of testing samples to be evaluated, y_i and $\hat{y}_i$ are the ground truth and the network prediction, respectively. An R^2 value close to 1 indicates higher prediction accuracy. SSIM is utilized to assess the similarity between two images. SSIM values range from 0 to 1, with higher values indicating better prediction accuracy. An SSIM value of 1 indicates identical images.

$$SSIM = \frac{1}{M} \sum_{m=1}^M \frac{(2\mu_{u,m}\mu_{v,m} + K_1)(2\sigma_{uv,m} + K_2)}{(\mu_{u,m}^2 + \mu_{v,m}^2 + K_1)(\sigma_{u,m}^2 + \sigma_{v,m}^2 + K_2)} \quad (12)$$

where u and v represent the true state variables and the predicted state variables, respectively. M is the number of local windows used to evaluate the images. μ and σ denote the mean and standard deviation, respectively. K_1 and K_2 are small constants to avoid division by zero.

4.2. Results of deep learning surrogate model

Figs. 5–7 display visualizations of the simulation results, prediction results, and errors for calcite, magnesite, and siderite, respectively, for a randomly selected case at time steps of 200 days, 250 days, 300 days, and 365 days. Over time, the accumulation of mineral precipitation becomes evident, with distinct differences among the minerals. Magnesite exhibits a smaller spatial distribution compared to other minerals—averaging 20–40 m less in extent across different scenarios discussed later—but shows a significantly greater precipitation volume,

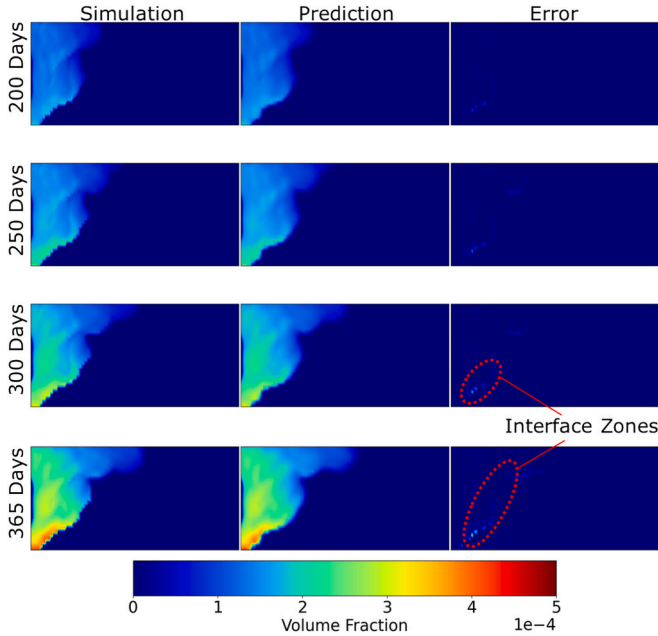


Fig. 5. Visualizations of the spatial distribution of calcite in the formation. The left column shows the simulation results, the middle column presents the predicted results from the surrogate model, and the right column displays the error values, which are the differences between the simulation and prediction. From top to bottom, each row corresponds to time steps at 200 days, 250 days, 300 days, and 365 days. The unit is volume fraction, representing the volumetric proportion of the mineral.

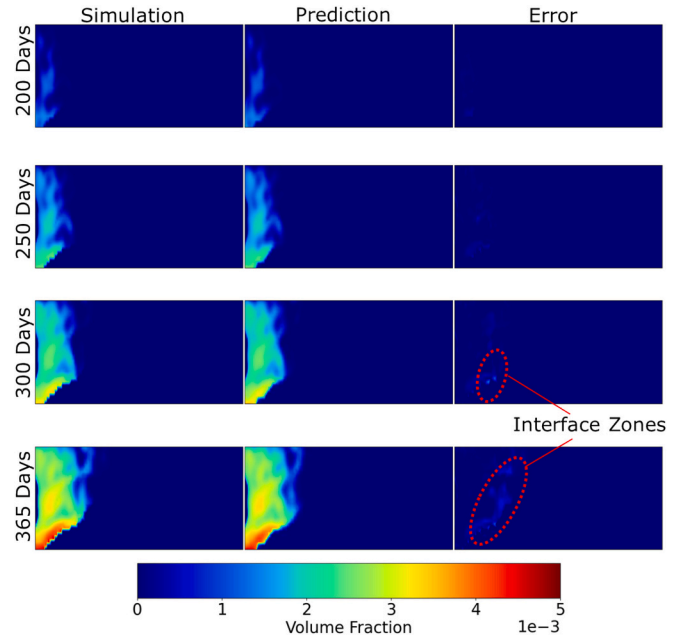


Fig. 6. Visualizations of the spatial distribution of magnesite in the formation. The left column shows the simulation results, the middle column presents the predicted results from the surrogate model, and the right column displays the error values, which are the differences between the simulation and prediction. From top to bottom, each row corresponds to time steps at 200 days, 250 days, 300 days, and 365 days. The unit is volume fraction, representing the volumetric proportion of the mineral.

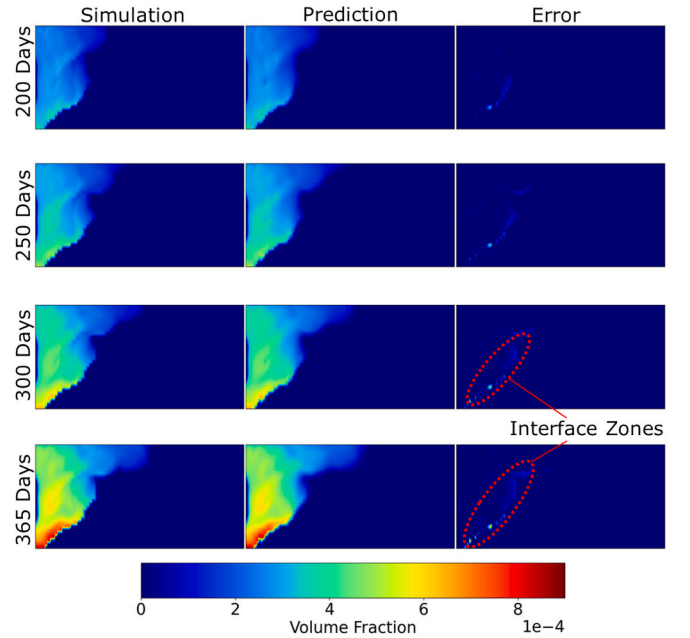


Fig. 7. Visualizations of the spatial distribution of siderite in the formation. The left column shows the simulation results, the middle column presents the predicted results from the surrogate model, and the right column displays the error values, which are the differences between the simulation and prediction. From top to bottom, each row corresponds to time steps at 200 days, 250 days, 300 days, and 365 days. The unit is volume fraction, representing the volumetric proportion of the mineral.

nearly an order of magnitude higher. In contrast, siderite and calcite display broader distribution patterns, with siderite having a greater precipitation volume than calcite. This disparity is attributed to the

faster dissolution rates of forsterite and fayalite, which release ions that enhance the carbonation capacity of both magnesite and siderite. These rapid dissolution kinetics reinforce the dominance of olivine-derived carbonates within the early simulation period, as captured by both the numerical and surrogate models. When comparing the predictions to the numerical simulation results, we observe that the neural network accurately predicts the spatial distribution of precipitation. It effectively captures the shapes and trends influenced by the heterogeneous permeability fields. The model also performs well in predicting the quantity of precipitation. However, the primary source of error arises at the interfaces between the precipitated zones and the unaffected zones. These regions experience rapid changes, making it challenging for the model to maintain high accuracy, leading to slight discrepancies in the predictions. To assist readers in identifying these areas, we have added dashed elliptical annotations in Fig., and to highlight the zones with the most pronounced errors.

Fig. 8 shows the stimulation results, prediction results, and errors for the increment of porosity for the same selected case at time steps of 200 days, 250 days, 300 days, and 365 days. The negative (−) sign designates a decrease in porosity, whereas the positive (+) sign indicates an increase. From the results, we can observe that as the CO₂ plume moves through the reservoir, the porosity initially increases and then decreases. At day 200, the entire reservoir appears red, indicating that the dissolution of initial minerals in the reservoir dominates. This phenomenon can enhance conductivity and facilitate the injection process. However, by day 300, we observe a decrease in porosity adjacent to the injection site, even lower than the original value, a process known as self-sealing. This occurs because, at this stage, precipitation becomes dominant over dissolution, leading to a reduction in porosity. Leveraging the porosity changes in the early stages of CO₂ injection could reduce injection difficulty by increasing porosity, while in the later stages, utilizing the self-sealing effect could reduce the risk of CO₂ leakage.

The prediction results also effectively capture the spatial structure of porosity changes, reflecting the heterogeneous permeability field's characteristics, particularly the irregularity of the changes. Notably, the

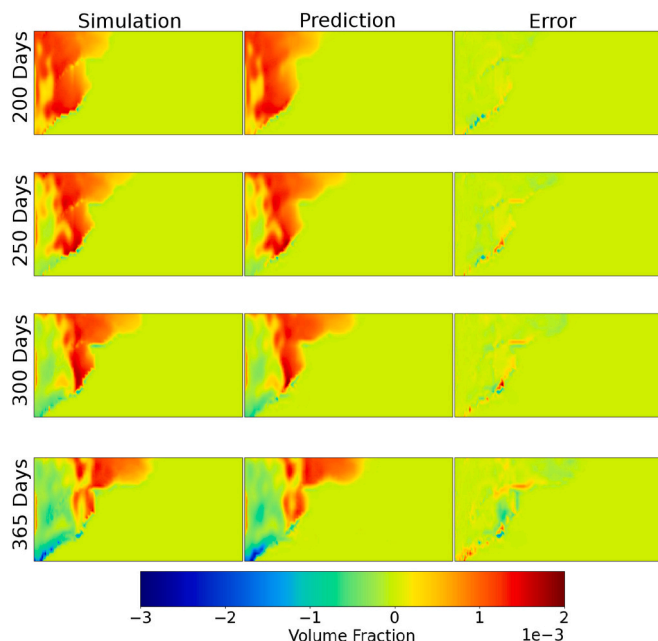


Fig. 8. Visualizations of porosity increment in the formation. The left column shows the simulation results, the middle column presents the predicted results from the surrogate model, and the right column displays the error values, which represent the differences between the simulation and prediction. From top to bottom, each row corresponds to time steps at 200 days, 250 days, 300 days, and 365 days.

neural network accurately predicts the trend of porosity changes over time, including the initial increase followed by a decrease, and captures the temporal sign changes of porosity increment. In terms of error analysis, discrepancies primarily occur at the interface between the affected and unaffected zones. This may be because the network's predictions tend to smooth functions, whereas the sharp changes at these interfaces reduce smoothness, leading to errors. Despite this, the network's overall performance in predicting porosity changes demonstrates its robustness and reliability in modelling complex geological processes.

4.3. Performance evaluation

Fig. 9a-d shows the temporal SSIM evolution of the output, providing a comprehensive view of the model's performance over time. As the time steps increase, the SSIM value tends to decrease. This is primarily because, during the early stages of CO₂ injection, the process is dominated by mineral dissolution, with almost no carbonate minerals being formed, and the effects are concentrated around the injection well. Therefore, the SSIM values are close to 1 due to the limited effect of the CO₂ injection in the early stages. However, as time progresses, the amount of precipitation increases, and the area affected by precipitation expands significantly. The balance between dissolution and precipitation in the site also begins to interact with the permeability field, resulting in a more uneven distribution of the minerals. This interaction introduces greater complexity into the system, challenging the model to maintain high prediction accuracy. Despite this, our network's overall prediction maintains an SSIM value greater than 0.95 at most time points, with an overall average SSIM greater than 0.96, indicating satisfactory performance. This demonstrates the model's robustness and reliability in capturing the evolving dynamics of CO₂ sequestration, even as the complexity of the system increases.

In the context of carbon sequestration, the accurate prediction of storage volume is of paramount importance. Efficient carbon sequestration not only relies on capturing and storing CO₂ but also on maximizing the storage efficiency within geological formations. High prediction accuracy for storage volume is crucial for ensuring the long-term stability and safety of the stored CO₂, as well as for optimizing the economic and operational aspects of sequestration projects. Given this, we evaluated the R² for storage volume prediction to assess the model's performance. Our calculation method involved multiplying the predicted and actual mineral volume fractions by the corresponding grid volumes for each grid cell and then summing them up to obtain the total volume of the stored mineral. Additionally, since the increment in porosity can have both positive and negative values, adding them together may not provide meaningful information. Therefore, we typically focus on the spatial distribution of porosity rather than calculating the R² for porosity volume.

We then plotted the R² values to validate the model's accuracy, as shown in Fig. 10a-c. The results indicate that the R² values for all minerals are greater than 0.98, demonstrating the high accuracy of our model in predicting storage volumes. This high degree of accuracy suggests that our model is robust and reliable for predicting the efficiency of carbon sequestration in geological formations. Moreover, with such a precise model, we can further utilize it to explore and identify optimal injection strategies. By simulating various injection scenarios, our model can help determine the most efficient methods for maximizing storage volume while minimizing risks associated with CO₂ leakage and ensuring the stability of the geological formation. This capability is vital for improving the overall efficacy of carbon sequestration projects and for developing best practices that can be applied in real-world scenarios.

Although the spatial distribution of gaseous and dissolved CO₂ is closely linked to mineral precipitation (Tariq et al., 2023), we did not include these variables as inputs in our surrogate model. This is because CO₂ distributions are themselves outputs of complex simulations; using them as inputs would require an additional prediction step, increasing

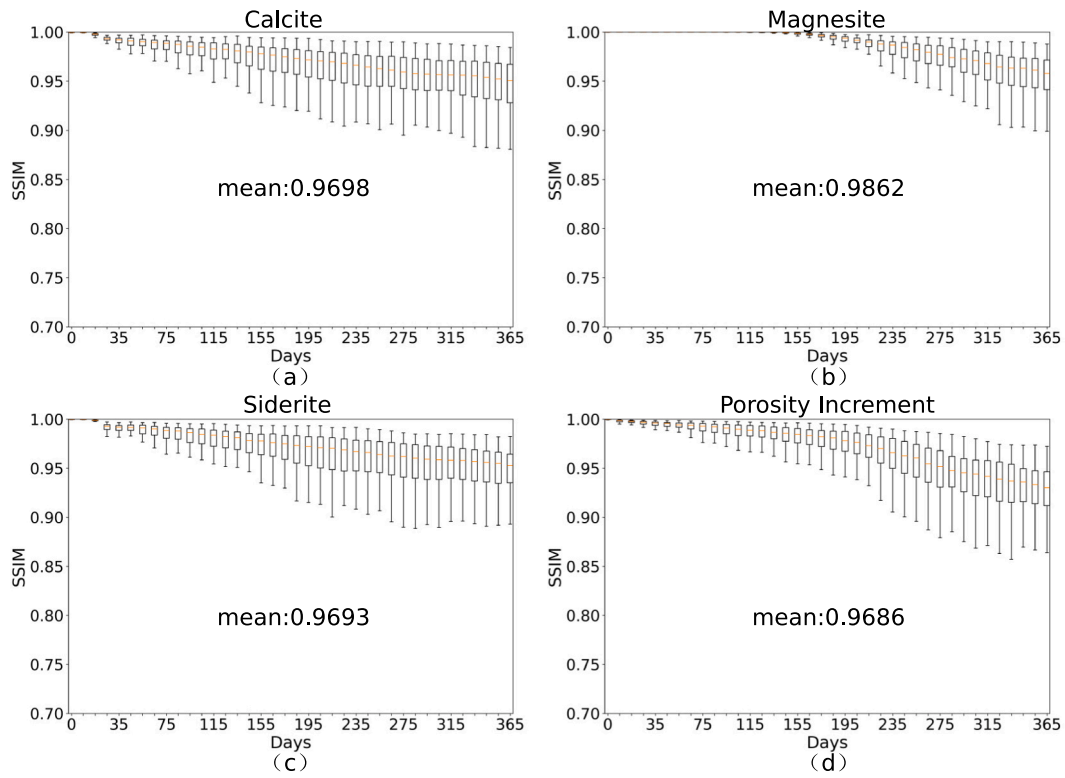


Fig. 9. Temporal SSIM boxplots comparing the prediction results and simulation results for: (a) calcite, (b) magnesite, (c) siderite, and (d) porosity increment.

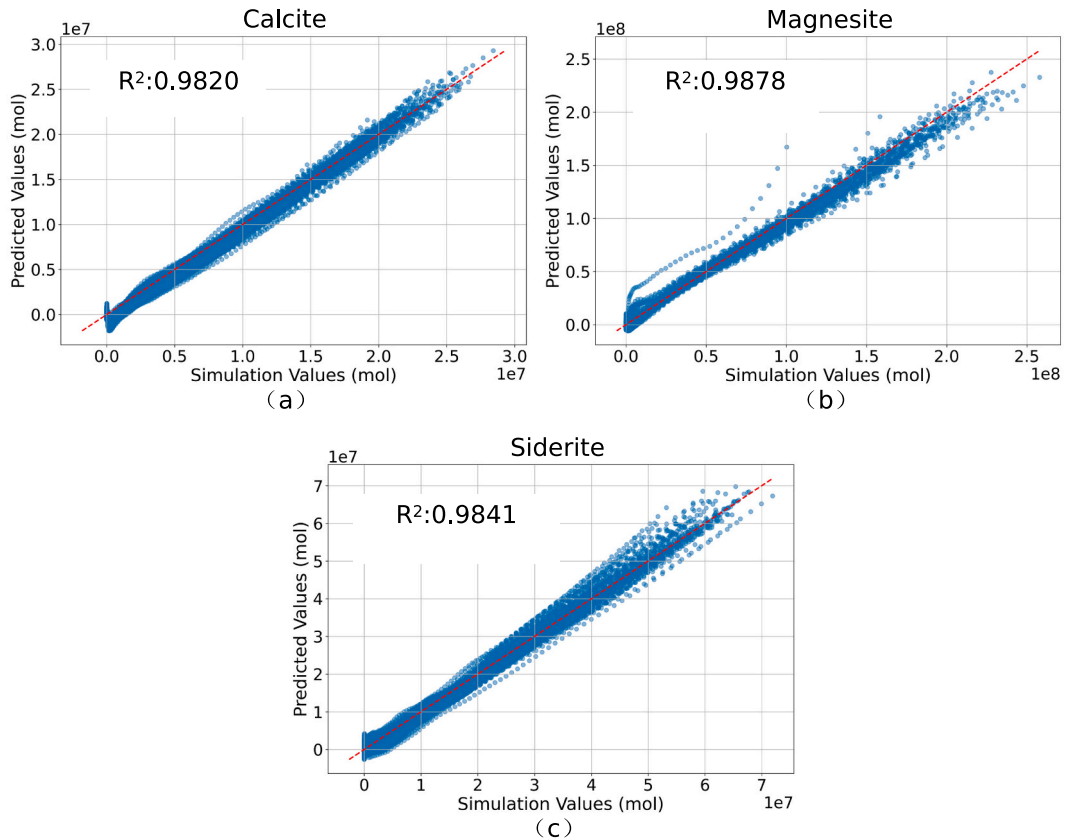


Fig. 10. Scatter plots comparing the total mineral quantities stored in the reservoir at all time points in the test dataset. The x-axis represents the simulation values, and the y-axis represents the prediction values from the surrogate model. The plots show comparisons for: (a) calcite, (b) magnesite, and (c) siderite.

computational cost. Instead, our model relies on fundamental inputs—such as permeability, porosity, injection conditions, and spatial-temporal coordinates—which implicitly capture the coupled behavior of CO₂ transport and geochemical reactions. We found this approach sufficiently accurate while maintaining model simplicity and generalizability across different scenarios.

5. Discussion

To evaluate the efficiency and accuracy of the surrogate model, we combined it with numerical simulations for parameter analysis. Specifically, we examined the effects of varying injection rates and locations on reservoir behavior by designing three distinct injection scenarios. The reservoir is divided into three sections: upper, middle, and lower, with each section measuring 33 m. In scenario 1 and scenario 2, injection rates of 3 kg/s and 5 kg/s are applied, respectively, with the injection location in the lower part of the reservoir. In scenario 3, the injection rate is kept at 5 kg/s, but the injection location is moved to the upper part of the reservoir, enabling a comparison with scenario 2 to assess the influence of different injection locations. For each scenario, 50 new cases were generated with different permeability maps to obtain statistical insights into the reservoir's response to CO₂ injection. The surrogate model demonstrated remarkable efficiency, requiring only 2.85 s to predict the outcomes for all 150 cases, compared to approximately 54,000 s needed by the numerical simulation. This highlights that the surrogate model is nearly 20,000 times faster than its numerical counterpart while maintaining high accuracy, even for unseen cases. The model's rapid and reliable predictions under diverse injection conditions make it an ideal tool for parameter analysis in basalt carbon sequestration.

5.1. Porosity evolution

We first compared the effects of different injection rates on porosity changes by selecting scenario 1 and scenario 2 for comparison. Two locations, 50 m and 150 m from the injection well were chosen to represent the near-wellbore and far-wellbore regions, respectively. The graphs showing the porosity variation over time at these locations are presented in Fig. 11a, b. The prediction results are depicted with dashed lines, while the simulation results are shown with solid lines. The surrogate model's predictions closely match the numerical simulations, effectively capturing the patterns of porosity changes under varying injection rates, providing valuable insights into the underlying trends.

In Fig. 11a, for both scenarios, the porosity increment at 50 m from the injection well initially increases, then decreases, showing a self-enhancing behavior near the well, followed by a self-limiting behavior

with porosity reduction near the injection zone. When comparing the two injection rates, it was observed that higher injection rates led to an earlier peak in porosity change, a greater increase in porosity, and an earlier transition into the self-limiting phase. The faster CO₂ injection creates a larger lateral pressure gradient, which accelerates the spread of CO₂ in the formation and increases CO₂ concentration. As CO₂ dissolves in water, more H⁺ ions are released, enhancing the dissolution of primary minerals, thus resulting in a higher peak porosity change. As mineral dissolution continues, cation concentrations in the solution increase. In cases with higher injection rates, cations accumulate more rapidly in the solution, leading to earlier and more substantial secondary precipitation. This results in a stronger self-limiting effect due to increased precipitation. Additionally, excess secondary precipitation can reduce mineral carbonation efficiency by hindering CO₂ contact with mineral surfaces. Under these conditions, porosity reduction around the injection well can severely limit the security and efficiency of ongoing CO₂ injection. Such porosity decreases may restrict injectivity, cause significant pressure build-ups around the wellbore, and potentially induce structural failures in the well construction. Fig. 11b compares the porosity changes at a location 150 m from the injection well. Notably, a temporal offset of approximately 50 days is evident: porosity changes begin around day 200 in Scenario 2, compared to day 250 in Scenario 1. In addition to the earlier onset, Scenario 2 also exhibits a steeper porosity evolution slope, reflecting the accelerated CO₂ plume migration under higher injection rates. However, due to the greater distance from the injection well, the CO₂ plume is just passing through this region, and the dissolution trend dominates. At this stage, the self-limiting effect has not yet occurred.

We selected Scenario 2 and Scenario 3 to compare the effects of different injection locations on porosity changes in the reservoir. Fig. 12 illustrates the spatial distribution of porosity under different injection locations. Fig. 12a shows the average porosity changes across the entire reservoir. We observe that the peak value in Scenario 3 is lower than in Scenario 2, and the peak in Scenario 3 appears further away compared to Scenario 2. When comparing Fig. 12b, we find that the peak values in both scenarios are approximately the same, differing only in location. This indicates that placing the injection well at the top of the reservoir both extends the affected area and concentrates it more in the upper part of the reservoir vertically. As a result, when averaging across the entire reservoir, the peak value decreases significantly.

To better demonstrate the impact of different injection locations on the vertical distribution of porosity in the reservoir, we averaged the porosity changes within 50 m of the wellbore and plotted the vertical porosity variation, as shown in Fig. 13a. We observed that mineral precipitation tends to occur at the bottom of the reservoir. For scenario 3, although the injection well is located at the top of the reservoir, the

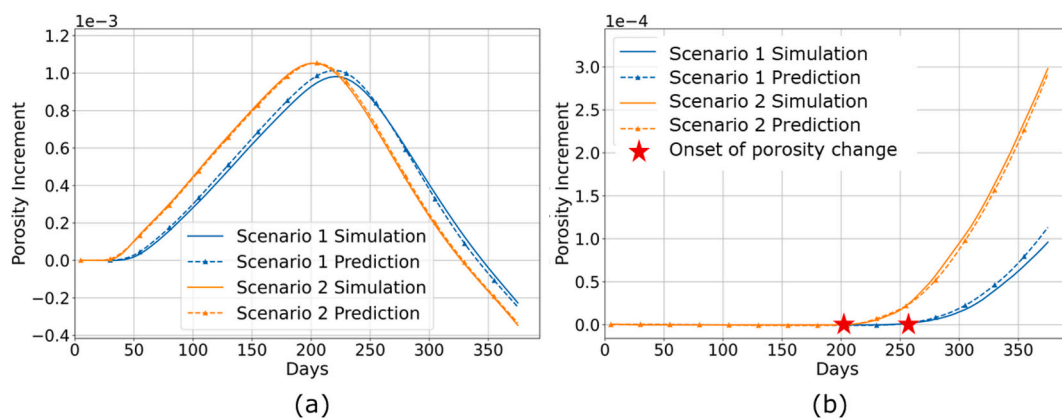


Fig. 11. Comparison of porosity changes over time at different locations in the reservoir for Scenario 1 and Scenario 2. (a) Comparison of simulation and prediction results of porosity increment at 50 m from the injection well. (b) Comparison of simulation and prediction results of porosity increment at 150 m from the injection well.

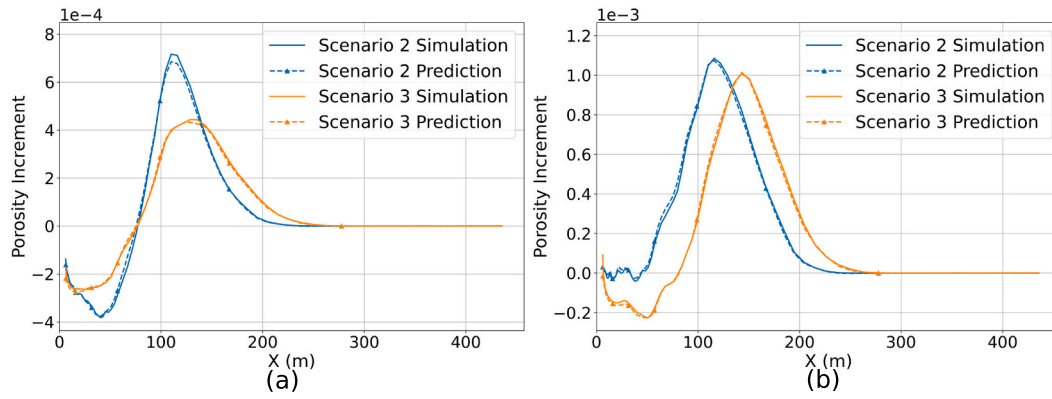


Fig. 12. Comparison of the average porosity changes at different distances from the injection well between scenario 2 and scenario 3 at the final time step (365 days). (a) Average porosity changes across the entire 100 m reservoir. (b) Average porosity changes in the upper 33 m section of the reservoir only.

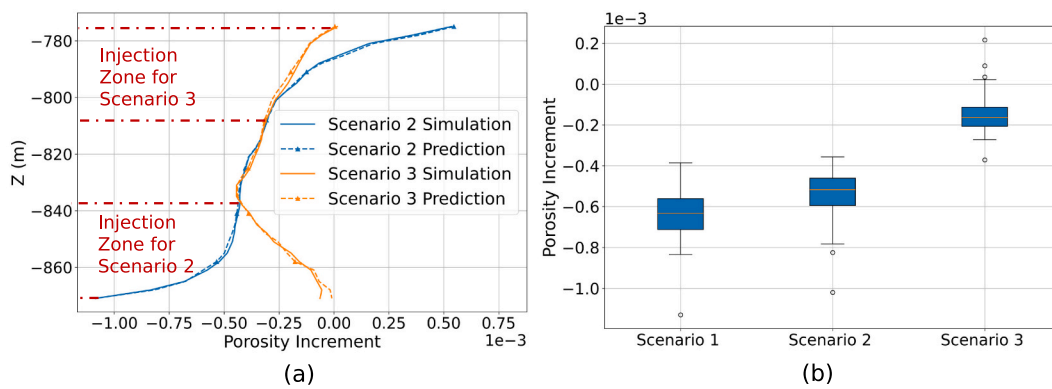


Fig. 13. Comparison of porosity changes near the wellbore (within 50 m from the injection location) at the final time step (365 days). (a) Average porosity changes around the wellbore plotted as a vertical porosity variation curve across the reservoir for scenario 2 and scenario 3. (b) Boxplot of porosity changes near the wellbore for all three scenarios.

minimum porosity is found slightly below the injection well. This could be due to the saturated CO_2 and dissolved metal cation solution, which, under the pressure gradient, spreads around the reservoir, including downward into the lower part. Additionally, the increased density of the solution could cause gravity-driven convection (Xu et al., 2024), leading to precipitation below the injection point. Although this results in reduced porosity, the peak decrease occurs beneath the injection well, thereby diminishing the sealing effectiveness. For scenario 2, with the injection well at the bottom of the reservoir, all minerals accumulate at the reservoir's base, causing a significant reduction in porosity near the injection well and leading to a severe self-sealing effect. This could result

in pressure buildup, which poses potential safety risks.

In Fig. 13b, we compared the average porosity changes near the injection well for all three scenarios. Overall, Scenario 3 demonstrates the smallest porosity reduction (-1.5×10^{-4}), while Scenarios 1 and 2 exhibit larger decreases (-6.4×10^{-4} and -5.4×10^{-4} , respectively). This indicates that the porosity loss near the injection well in Scenarios 1 and 2 is more than 3.5 times and 3 times greater than that in Scenario 3, respectively. From the perspective of mitigating pressure buildup and reducing self-sealing risks near the injection zone, these results suggest that placing the injection well at the top of the reservoir may offer better long-term injectivity and operational safety under the given conditions.

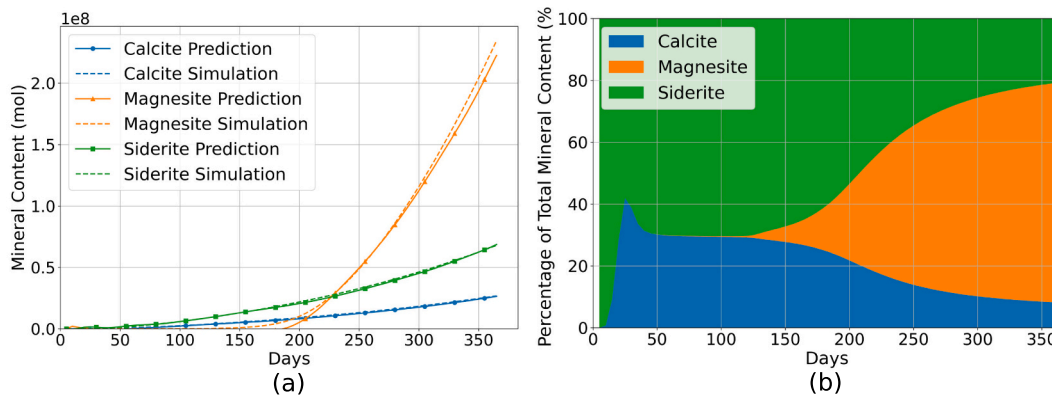


Fig. 14. Mineral content evolution in the reservoir for Scenario 2 over time. (a) Simulation and prediction results of the total amount of each mineral in the reservoir as a function of time. (b) Evolution of the percentage of each mineral content in the reservoir over time.

5.2. Mineral evolution and sequestration efficiency

The trends in mineral content and the percentage changes of individual minerals follow similar patterns across all three scenarios. As a result, we focus on scenario 2 as an example, shown in Fig. 14. In Fig. 14a, we observe that the simulation results align well with the prediction, and magnesite begins to precipitate significantly only after 150 days, eventually becoming the dominant mineral. However, due to the low precipitation amounts during the initial phase, fluctuations in the prediction can significantly affect the relative values. To better illustrate the trends, we have chosen to display only the simulation data. Fig. 14b further illustrates that during the initial phase of CO₂ injection, siderite is the primary mineral to precipitate, but it is later surpassed by magnesite. To enhance the interpretation of the precipitation sequence, we further present in Fig. 15 the evolution of key ion concentrations (Fe²⁺ and Mg²⁺) and pH over time. These results reveal that the lowest pH values are observed at the leading edge of the CO₂ plume, where CO₂ first dissolves into the formation water. In these acidic zones, Fe²⁺ concentrations reach their peak, creating favorable conditions for the early formation of siderite. This observation aligns with the findings of Menefee et al. (2017), who noted that under low-pH environments with available Fe²⁺, Mg²⁺, and Ca²⁺ ions, siderite is the most likely carbonate to precipitate. As the CO₂ front progresses and forsterite dissolves, the Mg²⁺ concentration increases, especially in the trailing region of the plume. This shift drives the delayed but dominant precipitation of magnesite, beginning around 100–250 days. Despite calcite precipitating relatively quickly, its overall amount remains low throughout the process. This is because the minerals in the original formation that can release calcium ions dissolve at a slower rate, which limits the formation of calcite and keeps its proportion consistently low during the injection period.

It is worth noting that magnesite is the dominant secondary mineral in our simulation results, whereas in some field studies—such as the Wallula pilot project—ankerite has been reported as a more common product. This discrepancy primarily arises from differences in the mineralogical composition and reaction duration. Specifically, the

simulated basalt sample used in our study has a relatively high olivine content (~13 %), in contrast to the much lower olivine content (0–3 %) observed at Wallula (Zakharova et al., 2012). Since olivine readily reacts with CO₂ to form magnesite (Chen et al., 2024b; Johnson et al., 2014), it becomes the dominant product under these simulated conditions.

Comparing the spatial distribution of minerals across different scenarios, we find similar patterns to those observed in porosity changes. In Fig. 16a, when comparing scenario 1 and scenario 2, the peak mineral content in scenario 2 is higher and located farther from the injection well than in scenario 1. In Fig. 16b, comparing scenario 2 and scenario 3, we see that the peak in scenario 3 is farther away from the well, but its magnitude is smaller than that of scenario 2. Fig. 16c compares the peak precipitation positions of various minerals across different scenarios, showing that magnesite precipitates closest to the well. Finally, Fig. 16d visually demonstrates that magnesite has the largest overall precipitation amount. This suggests that a significant portion of the CO₂ injected near the well is converted into magnesite precipitates. Therefore, we may need to pay closer attention to how these changes in mineral content near the injection well could impact long-term storage capacity.

Finally, we will discuss the issue of sequestration efficiency. We compared the sequestration efficiency across three different scenarios. Sequestration efficiency is defined as the molar amount of mineralized CO₂ divided by the molar amount of injected CO₂. Although our model adopts the geometric and hydrological parameters inspired by the Wallula pilot project, it does not use the actual mineral composition of the Wallula formation. Therefore, a direct comparison of mineralization efficiency with field results is not applicable. As shown in Fig. 17a, sequestration efficiency gradually increases over time. Initially, as CO₂ is injected into the reservoir, it must first dissolve into the formation water and subsequently dissolve sufficient minerals to release metal cations. Once these cations reach a critical concentration, mineral precipitation begins. Around day 150, the slope of the efficiency curve increases sharply, which coincides with the significant precipitation of magnesite, indicating that magnesite is the dominant mineral in these scenarios.

From Fig. 17b, we can see that when comparing scenarios 1 and 2, a

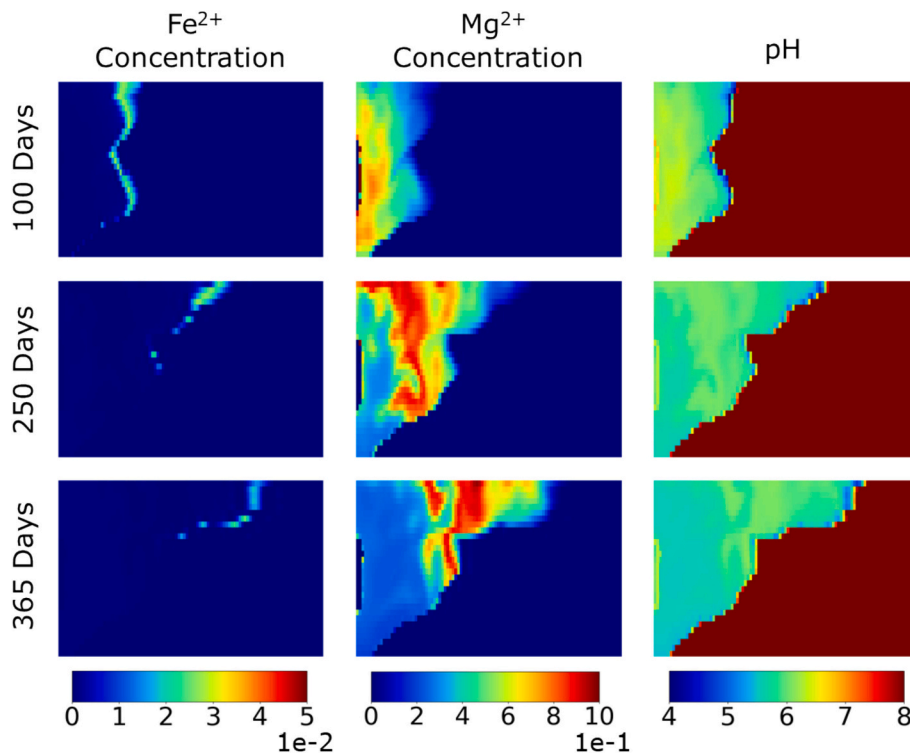


Fig. 15. Temporal evolution of Fe²⁺ and Mg²⁺ concentrations and pH during CO₂ injection.

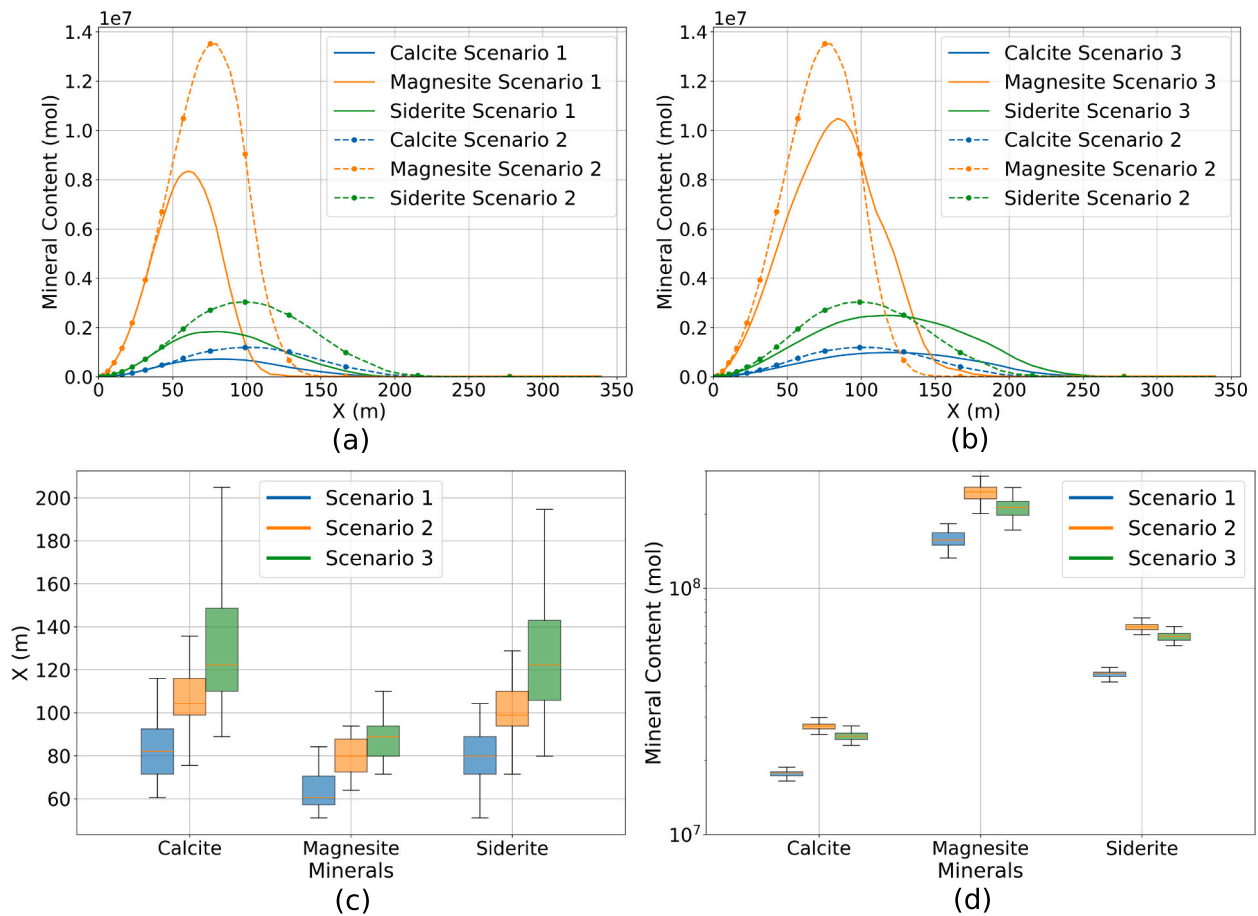


Fig. 16. Comparison of mineral content and peak positions at the final time step (365 days) across different scenarios. (a) Comparison of the spatial distribution of mineral content between scenario 1 and scenario 2. (b) Comparison of the spatial distribution of mineral content between scenario 2 and scenario 3. (c) Boxplot of the peak positions for precipitated minerals across all scenarios. (d) Boxplot comparing the total mineral content for each scenario.

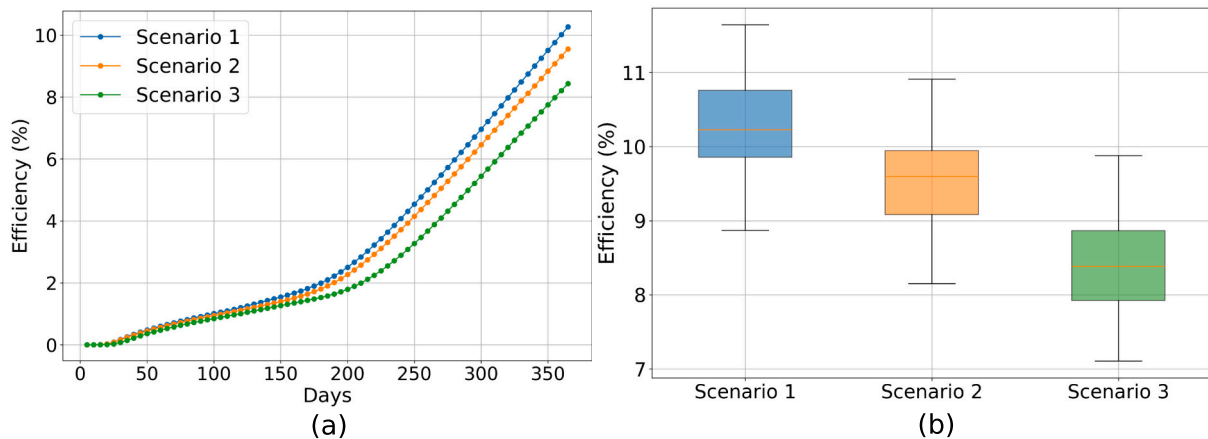


Fig. 17. Comparison of sequestration efficiency across different scenarios. (a) Curves showing the variation of sequestration efficiency over time for each scenario. (b) Boxplot comparing the final sequestration efficiency for each scenario.

slower injection rate results in higher sequestration efficiency—specifically, Scenario 1 is approximately 7.3 % higher than Scenario 2. This is because, with a smaller amount of CO_2 injected, a greater proportion of CO_2 can dissolve in the same volume of water, leading to a larger fraction of CO_2 being mineralized. When comparing Scenario 2 with Scenario 3, injecting CO_2 at the bottom of the reservoir results in 9.6 % sequestration efficiency, which is 14.3 % higher than the 8.4 % observed when injecting at the top. This improvement is likely due to buoyancy

effects: CO_2 tends to rise due to its lower density compared to formation water. If the injection well is located at the top of the reservoir, the CO_2 will mostly remain in the upper portion, minimizing its interaction with water and minerals in the lower part of the reservoir. In contrast, when the CO_2 is injected at the bottom, the upward-migrating plume will pass through the water and minerals in the lower and middle parts of the reservoir, promoting more extensive dissolution and mineralization along its migration path. Thus, in terms of maximizing dissolution and

Table 5Effects of injection rate and location on porosity and CO₂ sequestration efficiency.

Scenario	Injection Location	Injection Rate	Porosity Changes Near Wellbore	Sequestration Efficiency
1	Bottom	3 kg/s	-6.4×10^{-4}	10.3 %
2	Bottom	5 kg/s	-5.4×10^{-4}	9.6 %
3	Top	5 kg/s	-1.5×10^{-4}	8.4 %

mineralization efficiency, it may be preferable to place the injection well at the bottom of the reservoir. However, when considering pressure buildup and porosity reduction, placing the well at the top could help mitigate self-sealing effects and related injection issues. A comparative summary of these trade-offs across different scenarios is presented in Table 5. Beyond injection rate and vertical placement, other reservoir characteristics and operational constraints must also be considered to determine the optimal well configuration in practice. The distribution of permeability in the reservoir plays a crucial role in determining the optimal injection location. High-permeability layers enhance injectivity and facilitate CO₂ migration, while low-permeability layers may contribute to pressure buildup and potential leakage risks (Jun et al., 2019). Moreover, the choice between vertical and horizontal wells is influenced by pressure constraints and economic considerations. Horizontal wells can accommodate higher injection rates in pressure-limited settings, although they come with higher drilling costs (Okwen et al., 2011). Finally, the presence of faults and the associated stress regime should also be considered. For example, in normal fault stress regimes, increased horizontal stress can stabilize the reservoir, whereas in reverse fault regimes, it may elevate shear stress and induce failure (Vilarrasa, 2014). These additional factors are important for guiding safe and efficient well design in practical CCS operations.

6. Conclusions

This study firstly establishes a comprehensive numerical framework for simulating CO₂ mineral sequestration in basalt formations, incorporating multiphase flow and complex geochemical reactions involving 11 primary and secondary minerals. The model also accounts for dynamic changes in porosity and permeability due to mineral precipitation and dissolution, offering a more realistic representation of reservoir evolution during CO₂ injection.

Building upon this high-fidelity simulation framework, we developed a deep learning surrogate model based on an attention-augmented U-Net architecture. The model takes as input both reservoir characteristics (e.g., permeability, porosity, pore volume, and spatial coordinates) and injection parameters (e.g., rate and location) to rapidly predict mineral distributions and porosity evolution across time. Unlike conventional workflows that require intermediate predictions such as CO₂ saturation, our model enables direct, one-step forecasting of mineralization outcomes. Moreover, it implicitly accounts for the dynamic feedback between porosity-permeability variations and CO₂ flow, enhancing the realism of the predictions. Compared to traditional simulations, the surrogate model achieves a speedup of approximately 20,000 times while maintaining high accuracy (SSIM > 0.96, R² > 0.98), making it a powerful tool for large-scale parameter studies and optimization.

Appendix A. Model validation

To validate our numerical algorithm and parameter selection, we referred to static batch experiment and numerical benchmark conducted by Xiong et al. (2017). The experiment used cylindrical cores (25.4 mm diameter, 43 mm length) that were cut into half cylinders, with an 11 mm wide groove milled on one surface (Fig. A1a). We developed a similar 1D reactive transport model focusing on the trapping rate and spatial distribution of siderite, as shown in Fig. A1b. We divided the basalt core into 76 grids to capture the distribution of carbonate diffusion and carbonate precipitation during CO₂ infiltration. To align with the experimental conditions in the batch reactors, a constant pressure boundary, consisting of pure water

Leveraging the surrogate model, we performed a parametric analysis of three injection scenarios. The results show that reducing the injection rate increases the proportion of CO₂ mineralized by enhancing dissolution efficiency. Injecting CO₂ at the bottom of the reservoir further improves sequestration performance by promoting upward migration and wider interaction with reactive minerals. However, placing the injection well at the top significantly reduces porosity loss near the wellbore—by more than 70 % compared to other scenarios—thereby mitigating pressure buildup and the risk of self-sealing. Among the precipitated carbonates, magnesite emerges as the most dominant phase, both in quantity and spatial concentration, suggesting its critical role in long-term CO₂ storage.

Future work can focus on implementing variable-size grids to allow for more precise modelling of heterogeneous geological formations, leading to more accurate predictions of CO₂ sequestration dynamics. Additionally, considering the interaction between multiple wells in extraction and injection scenarios can provide a comprehensive understanding of system dynamics, optimizing the placement and operation of multiple wells. In the current study, we limited the mineral system to match experimental constraints and focus on early-stage dominant reactions; future extensions will aim to overcome this simplification. A key direction for improvement lies in incorporating more complex geochemical reaction networks into the model. This includes accounting for slower-reacting primary minerals (e.g., pyroxene, plagioclase) and a broader suite of secondary mineral phases such as clay minerals, dolomite, and ankerite. To address long-term prediction uncertainty, we plan to develop extrapolation-capable surrogate models trained on extended multi-year simulation datasets. This will enable validation beyond the current one-year scope and improve confidence in long-term CO₂ sequestration forecasts. Finally, the model can be used to solve optimization problems, determining the optimal injection strategies and locations to maximize CO₂ sequestration while minimizing risks and costs.

CRedit authorship contribution statement

WeiQuan Ouyang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Zhao Feng:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology. **Fengshou Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Zhao Xia:** Writing – review & editing, Validation. **Xianda Shen:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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equilibrated with CO₂ at 10 MPa and the experimental temperature (100 °C), was imposed at the inlet. The ion concentrations at the inlet were set using PHREEQC (Parkhurst and Appelo, 1999). A no-flux boundary was set at the bottom of the fracture for the sealed end of the core in the experiment. Diffusion was the only transport mechanism considered, with fixed diffusion coefficients of $7.5 \times 10^{-9} \text{ m}^2/\text{s}$ (100 °C) applied for all species (Cadogan et al., 2014).

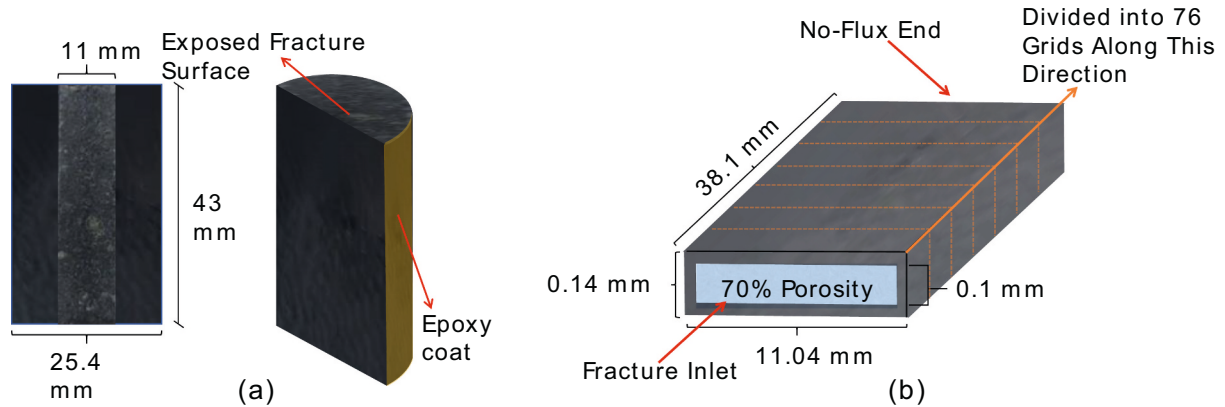


Fig. A1. (a) Conceptual diagram of a fractured basalt core (modified from Adeoye et al. (2017)). (b) Conceptual schematic of the numerical model (modified from Xiong et al. (2017)).

Fig. A2a and Fig. A2b shows the comparison of siderite distribution at 20 weeks and 40 weeks between our TOUGHREACT simulations and the benchmarked simulation results from Xiong et al. (2017). First, looking at the results after 40 weeks, the spatial distribution and numerical values of siderite predicted by our model closely match the trends in benchmarked data. The predicted siderite formation along the fracture follows similar spatial distribution trends as observed in the experiment. Near the fracture inlet, the low pH and low concentration of divalent cations in the bulk solution limit the formation of carbonates, including siderite (Giammar et al., 2014). The predicted amount of siderite increases sharply after the dissolution-only region, reaching a maximum around 10–20 mm from the fracture inlet. In the maximum precipitation zone, there was sufficient space for solutes to diffuse through the fluid in the fracture, resulting in significant siderite accumulation even beyond this region. The comparison at 20 weeks shows a similar pattern to that at 40 weeks, with minimal precipitation near the inlet and a peak in precipitation at a subsequent position, followed by a decrease after the peak. However, due to less extensive diffusion of carbonate ions at the 20-week, the location of maximum precipitation is closer to the inlet compared to the 40-week results. To quantify the agreement, we calculated the coefficient of determination (R^2) between our simulation results and the digitized benchmark profiles. The R^2 values exceed 0.9 for both time steps, indicating a high level of consistency and validating the accuracy of our model.

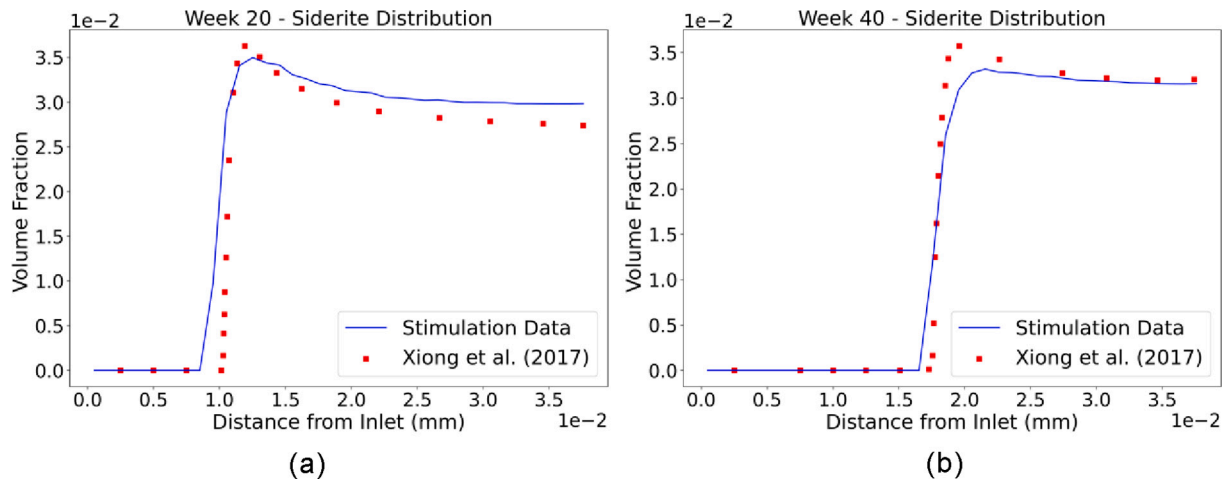


Fig. A2. (a) Comparison of siderite volume fraction distribution along the distance from the inlet after 20 weeks. (b) Comparison of siderite volume fraction distribution along the distance from the inlet after 40 weeks. The blue line represents the simulation data from our model, while the red squares represent the simulation data benchmarked by Xiong et al. (2017). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To verify grid independence, we tested five mesh configurations with 30 vertical cells and 25–200 radial cells. CO₂ saturation distribution and computational time were used to evaluate convergence. As shown in Fig. A3, results stabilized beyond 100 radial cells, while computational cost continued to rise. Thus, the 30×100 grids offer a good balance between accuracy and efficiency.

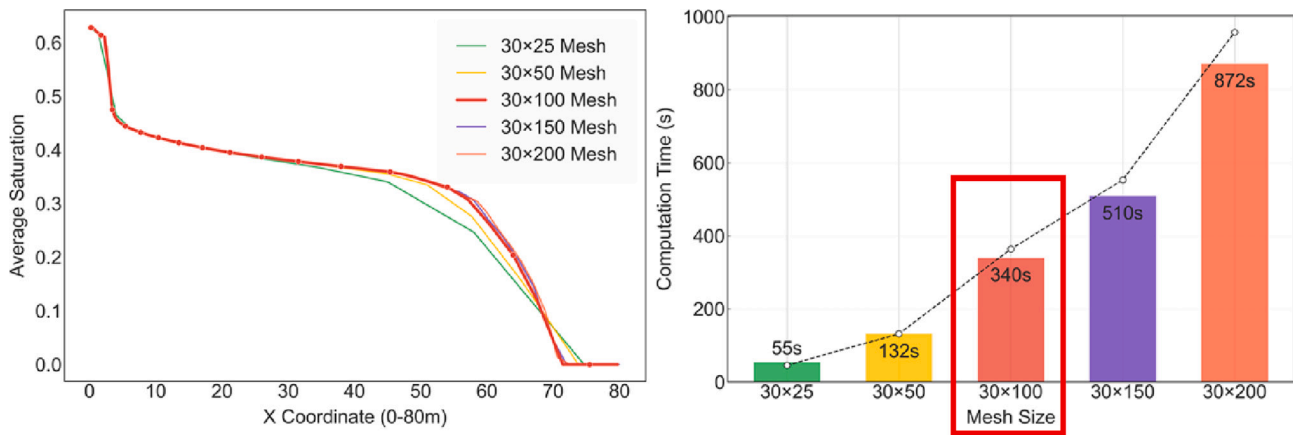


Fig. A3. Mesh independence verification results. (a) Average saturation distribution along the 0–80 m domain. (b) Computational time for different grid resolutions”.

Appendix B. Deep learning models comparison

We compared the original U-Net model with the U-Net with Attention model used in the study. The architecture of the original U-Net model is presented in Table B.1. We used SSIM and R^2 as comparison metrics, with the final results shown in Table B.2. The table shows that the SSIM values of each output feature are approximately 1 % higher for the U-Net with Attention compared to the original U-Net. Although the R^2 value for calcite in the U-Net with Attention is slightly lower than that of the original U-Net, the values for the other two minerals are significantly higher. Overall, we can say that the U-Net with Attention demonstrates greater stability and accuracy than the original U-Net.

The improved performance of the U-Net with Attention model can be attributed to the enhanced global perspective provided by the attention mechanism. While CNNs are highly effective at capturing local features, they may overlook broader contextual information due to their limited receptive fields. By integrating a transformer-based attention mechanism into the U-Net, particularly at the deeper layers of the network, the model gains the ability to focus on relevant global features across the entire input space. This global awareness allows the U-Net with Attention to better capture complex spatial relationships and dependencies, resulting in improved performance in terms of both stability and accuracy.

Table B.1
Summary of the original U-Net model architecture.

Name	Type	Output shape
Input		7, 30, 100
Encoder1	Conv2d/Conv2d/MaxPool2d	16, 15, 50
Encoder2	Conv2d/Conv2d/MaxPool2d	32, 7, 25
Encoder3	Conv2d/Conv2d/MaxPool2d	64, 3, 12
Encoder4	Conv2d/Conv2d/MaxPool2d	128, 1, 6
Conv	Conv2d/Conv2d	256, 1, 6
Decoder1	TransposeConv2d/Conv2d/Conv2d	128, 3, 12
Decoder2	TransposeConv2d/Conv2d/Conv2d	64, 7, 25
Decoder3	TransposeConv2d/Conv2d/Conv2d	32, 15, 50
Decoder4	TransposeConv2d/Conv2d/Conv2d	16, 30, 100
Conv	1 × 1Conv2d	1, 30, 100
Output		1, 30, 100

Table B.2
Performance comparison of U-Net with attention model and original U-Net model.

	SSIM		R^2	
	U-Net with attention	U-Net	U-Net with attention	U-Net
Magnesite	0.9862	0.9829	0.9878	0.9728
Porosity increment	0.9686	0.9585		
Calcite	0.9698	0.9637	0.9820	0.9883
Siderite	0.9693	0.9582	0.9841	0.9615

Data availability

Data will be made available on request.

References

Adeoye, J.T., Menefee, A.H., Xiong, W., Wells, R.K., Skemer, P., Giammar, D.E., Ellis, B. R., 2017. Effect of transport limitations and fluid properties on reaction products in

- fractures of unaltered and serpentinized basalt exposed to high PCO fluids. *Int. J. Greenh. Gas Control.* 63, 310–320. <https://doi.org/10.1016/j.ijggc.2017.06.003>.
- Bachu, S., 2008. CO₂ storage in geological media: Role, means, status and barriers to deployment. *Prog. Energy Combust. Sci.* 34, 254–273. <https://doi.org/10.1016/j.pecc.2007.10.001>.
- Banfield, J.F., Jones, B.F., Veblen, D.R., 1991. An AEM-TEM study of weathering and diagenesis, Abert Lake, Oregon: I. Weathering reactions in the volcanics. *Geochim. Cosmochim. Acta* 55, 2781–2793.
- Blunden, J., Arndt, D.S., 2019. State of the climate in 2018. *Bull. Am. Meteorol. Soc.* 100, Si-S306. <https://doi.org/10.1175/2019BAMSStateoftheClimate.1>.
- Cadogan, S.P., Maitland, G.C., Trusler, J.P.M., 2014. Diffusion Coefficients of CO₂ and N₂ in Water at Temperatures between 298.15 K and 423.15 K at pressures up to 45 MPa. *J. Chem. Eng. Data* 59, 519–525. <https://doi.org/10.1021/je401008s>.
- Carroll, S.A., McNab, W.W., Dai, Z., Torres, S.C., 2013. Reactivity of Mount Simon Sandstone and the Eau Claire Shale under CO₂ Storage Conditions. *Environ. Sci. Technol.* 47, 252–261. <https://doi.org/10.1021/es301269k>.
- Chen, B., Harp, D.R., Lu, Z., Pawar, R.J., 2020. Reducing uncertainty in geologic CO₂ sequestration risk assessment by assimilating monitoring data. *Int. J. Greenh. Gas Control.* 94, 102926. <https://doi.org/10.1016/j.ijggc.2019.102926>.
- Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A.L., Zhou, Y., 2021. TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation. <https://doi.org/10.48550/ARXIV.2102.04306>.
- Chen, B., Li, Q., Tan, Y., Zhang, Y., Yu, T., Ma, J., Zhong, Y., Li, X., 2024a. Caprock sealing integrity and key indicators of CO₂ geological storage considering the effect of hydraulic-mechanical coupling: X field in the Bohai Bay Basin, China. *Eng. Geol.* 342, 107741. <https://doi.org/10.1016/j.enggeo.2024.107741>.
- Chen, Y., Seyyedi, M., Clennell, B., 2024b. Petrophysical recipe for in-situ CO₂ mineralization in basalt rocks. *Adv. Geo-Energy Res.* 11, 152–160. <https://doi.org/10.46690/ager.2024.02.07>.
- Civan, F., 2001. Scale effect on porosity and permeability: Kinetics, model, and correlation. *AIChE J.* 47, 271–287. <https://doi.org/10.1002/aic.690470206>.
- Craig, D.C., Loughnan, F.C., 1964. Chemical and mineralogical transformations accompanying the weathering of basic volcanic rocks from New South Wales. *Soil Res.* 2, 218–234. <https://doi.org/10.1071/SR9640218>.
- Czarnecki, W.M., Osindero, S., Jaderberg, M., Swirszcz, G., Pascanu, R., 2017. Sobolev training for neural networks. In: *Advances in Neural Information Processing Systems*. Curran Associates, Inc.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N., 2020. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. <https://doi.org/10.48550/ARXIV.2010.11929>.
- Dou, W., Geng, C., Lin, M., Jiang, W., Ji, L., Cao, G., 2024. A Pore-Scale phase Field Model for CO₂–Fluid–Basalt Interactions. *ACS Omega* 9, 28648–28658. <https://doi.org/10.1021/acsomega.4c02707>.
- Doughty, C., 2010. Investigation of CO₂ plume behavior for a large-scale pilot test of geologic carbon storage in a saline formation. *Transp. Porous Media* 82, 49–76. <https://doi.org/10.1007/s11242-009-9396-z>.
- Feng, Z., Tariq, Z., Shen, X., Yan, B., Tang, X., Zhang, F., 2024. An encoder-decoder ConvLSTM surrogate model for simulating geological CO₂ sequestration with dynamic well controls. *Gas Sci. Eng.* 125, 205314. <https://doi.org/10.1016/j.jgsce.2024.205314>.
- Giammar, D.E., Wang, F., Guo, B., Surface, J.A., Peters, C.A., Conradi, M.S., Hayes, S.E., 2014. Impacts of Diffusive Transport on Carbonate Mineral Formation from Magnesium Silicate-CO₂-Water Reactions. *Environ. Sci. Technol.* 48, 14344–14351. <https://doi.org/10.1021/es504047t>.
- Golubev, S.V., Bénézet, P., Schott, J., Dandurand, J.L., Castillo, A., 2009. Siderite dissolution kinetics in acidic aqueous solutions from 25 to 100 °C and 0 to 50 atm pCO₂. *Chem. Geol.* 265, 13–19. <https://doi.org/10.1016/j.chemgeo.2008.12.031>.
- Gudbrandsson, S., Wolff-Boenisch, D., Gislason, S.R., Oelkers, E.H., 2011. An experimental study of crystalline basalt dissolution from 2 ≤ pH ≤ 11 and temperatures from 5 to 75 °C. *Geochim. Cosmochim. Acta* 75, 5496–5509. <https://doi.org/10.1016/j.gca.2011.06.035>.
- Jayne, R.S., Wu, H., Pollyea, R.M., 2019. Geologic CO₂ sequestration and permeability uncertainty in a highly heterogeneous reservoir. *Int. J. Greenh. Gas Control.* 83, 128–139. <https://doi.org/10.1016/j.ijggc.2019.02.001>.
- Jiang, Z., Zhu, M., Lu, L., 2024. Fourier-MIONet: Fourier-enhanced multiple-input neural operators for multiphase modeling of geological carbon sequestration. *Reliab. Eng. Syst. Saf.* 251, 110392. <https://doi.org/10.1016/j.ress.2024.110392>.
- Johnson, N.C., Thomas, B., Maher, K., Rosenbauer, R.J., Bird, D., Brown, G.E., 2014. Olivine dissolution and carbonation under conditions relevant for in situ carbon storage. *Chem. Geol.* 373, 93–105. <https://doi.org/10.1016/j.chemgeo.2014.02.026>.
- Jun, C., Kim, M., Shin, H., 2019. Optimization of well placement and operating conditions for various well patterns in CO₂ sequestration in the Pohang Basin, Korea. *Int. J. Greenh. Gas Control.* 90, 102810. <https://doi.org/10.1016/j.ijggc.2019.102810>.
- Kelemen, P.B., Matter, J., 2008. In situ carbonation of peridotite for CO₂ storage. *Proc. Natl. Acad. Sci. USA* 105, 17295–17300. <https://doi.org/10.1073/pnas.0805794105>.
- Kitanidis, P.K., 2015. Persistent questions of heterogeneity, uncertainty, and scale in subsurface flow and transport. *Water Resour. Res.* 51, 5888–5904. <https://doi.org/10.1002/2015WR017639>.
- Knaus, K.G., Johnson, J.W., Steefel, C.I., 2005. Evaluation of the impact of CO₂, CO-contaminant gas, aqueous fluid and reservoir rock interactions on the geologic sequestration of CO₂. *Chem. Geol.* 217, 339–350. <https://doi.org/10.1016/j.chemgeo.2004.12.017>.
- Lasaga, A.C., Soler, J.M., Ganor, J., Burch, T.E., Nagy, K.L., 1994. Chemical weathering rate laws and global geochemical cycles. *Geochim. Cosmochim. Acta* 58, 2361–2386. [https://doi.org/10.1016/0016-7037\(94\)90016-7](https://doi.org/10.1016/0016-7037(94)90016-7).
- Liu, D., Agarwal, R., Li, Y., Yang, S., 2019. Reactive transport modeling of mineral carbonation in unaltered and altered basalts during CO₂ sequestration. *Int. J. Greenh. Gas Control.* 85, 109–120. <https://doi.org/10.1016/j.ijggc.2019.04.006>.
- Luquot, L., Gouze, P., 2009. Experimental determination of porosity and permeability changes induced by injection of CO₂ into carbonate rocks. *Chem. Geol.* 265, 148–159. <https://doi.org/10.1016/j.chemgeo.2009.03.028>.
- Matter, J.M., Stute, M., Snæbjörnsdóttir, S.O., Oelkers, E.H., Gislason, S.R., Aradóttir, E. S., Sigfusson, B., Gunnarsson, I., Sigurdardóttir, H., Gunnlaugsson, E., Axelsson, G., Alfredsson, H.A., Wolff-Boenisch, D., Mesfin, K., Taya, D.F.D.L.R., Hall, J., Dideriksen, K., Broecker, W.S., 2016. Rapid carbon mineralization for permanent disposal of anthropogenic carbon dioxide emissions. *Science* 352, 1312–1314. <https://doi.org/10.1126/science.aad8132>.
- McGrail, B.P., Schaefer, H.T., Ho, A.M., Chien, Y., Dooley, J.J., Davidson, C.L., 2006. Potential for carbon dioxide sequestration in flood basalts. *J. Geophys. Res.* 111, 2005JB004169. <https://doi.org/10.1029/2005JB004169>.
- McGrail, B.P., Sullivan, E.C., Spang, F.A., Bacon, D.H., Hund, G., Thorne, P.D., Thompson, C.J., Reidel, S.P., Colwell, F.S., 2009. Preliminary Hydrogeologic Characterization Results from the Wallula Basalt Pilot Study. Montana State Univ, Bozeman, MT (United States).
- McGrail, B.P., Schaefer, H.T., Spang, F.A., Cliff, J.B., Qafoku, O., Horner, J.A., Thompson, C.J., Owen, A.T., Sullivan, C.E., 2017. Field Validation of Supercritical CO₂ Reactivity with Basalts. *Environ. Sci. Technol. Lett.* 4, 6–10. <https://doi.org/10.1021/acs.estlett.6b00387>.
- Menefee, A.H., Li, P., Giammar, D.E., Ellis, B.R., 2017. Roles of transport limitations and mineral heterogeneity in carbonation of fractured basalts. *Environ. Sci. Technol.* 51, 9352–9362. <https://doi.org/10.1021/acs.est.7b00326>.
- Müller, S., Schüler, L., Zech, A., Heße, F., 2022. GSTools v1.3: a toolbox for geostatistical modelling in Python. *Geosci. Model Dev.* 15, 3161–3182. <https://doi.org/10.5194/gmd-15-3161-2022>.
- Oelkers, E.H., Arkadakskiy, S., Afifi, A.M., Hoteit, H., Richards, M., Fedorik, J., Delaunay, A., Torres, J.E., Ahmed, Z.T., Kunnummal, N., Gislason, S.R., 2022. The subsurface carbonation potential of basaltic rocks from the Jizan region of Southwest Saudi Arabia. *Int. J. Greenh. Gas Control.* 120, 103772. <https://doi.org/10.1016/j.ijggc.2022.103772>.
- Okwen, R., Stewart, M., Cunningham, J., 2011. Effect of well Orientation (Vertical vs. Horizontal) and well Length on the Injection of CO₂ in Deep Saline Aquifers. *Transp. Porous Media* 90, 219–232. <https://doi.org/10.1007/s11242-010-9686-5>.
- Palandri, J.L., Kharaka, Y.K., 2004. A Compilation of Rate Parameters of Water-Mineral Interaction Kinetics for Application to Geochemical Modeling. US Geological Survey.
- Parkhurst, D.L., Appelo, C.A.J., 1999. User's Guide to PHREEQC (Version 2): A Computer Program for Speciation, Batch-Reaction, One-Dimensional Transport, and Inverse Geochemical Calculations. US Geological Survey.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimeshin, N., Antiga, L., 2019. Pytorch: an imperative style, high-performance deep learning library. *Adv. Neural Inf. Process. Syst.* 32.
- Pokrovsky, O.S., Schott, J., 2002. Surface chemistry and dissolution kinetics of divalent metal carbonates. *Environ. Sci. Technol.* 36, 426–432. <https://doi.org/10.1021/es010925u>.
- Pokrovsky, O.S., Golubev, S.V., Schott, J., 2005. Dissolution kinetics of calcite, dolomite and magnesite at 25 °C and 0 to 50 atm pCO₂. *Chem. Geol.* 217, 239–255. <https://doi.org/10.1016/j.chemgeo.2004.12.012>.
- Pollyea, R.M., 2016. Influence of relative permeability on injection pressure and plume configuration during CO₂ injections in a mafic reservoir. *Int. J. Greenh. Gas Control.* 46, 7–17. <https://doi.org/10.1016/j.ijggc.2015.12.025>.
- Pruess, K., 1991. A General Purpose Numerical Simulator for Multiphase Fluid and Heat. LBL-29400, Berkeley, California, USA.
- Reed, M.H., 1982. Calculation of multicomponent chemical equilibria and reaction processes in systems involving minerals, gases and an aqueous phase. *Geochim. Cosmochim. Acta* 46, 513–528. [https://doi.org/10.1016/0016-7037\(82\)90155-7](https://doi.org/10.1016/0016-7037(82)90155-7).
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B., 2022. High-resolution image synthesis with latent diffusion models. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10684–10695.
- Rudnick, R.L., White, W.M., 2018. In: White, W.M. (Ed.), *Earth's continental crust. Encyclopedia of Geochemistry: A Comprehensive Reference Source on the Chemistry of the Earth*, pp. 392–418. https://doi.org/10.1007/978-3-319-39312-4_277.
- Snæbjörnsdóttir, S.O., Sigfusson, B., Marieni, C., Goldberg, D., Gislason, S.R., Oelkers, E. H., 2020. Carbon dioxide storage through mineral carbonation. *Nat. Rev. Earth Environ.* 1, 90–102. <https://doi.org/10.1038/s43017-019-0011-8>.
- Steeff, C.I., Lasaga, A.C., 1994. A coupled model for transport of multiple chemical species and kinetic precipitation/dissolution reactions with application to reactive flow in single phase hydrothermal systems. *Am. J. Sci.* 294, 529–592. <https://doi.org/10.2475/ajs.294.5.529>.
- Tariq, Z., Yildirim, E.U., Gudala, M., Yan, B., Sun, S., Hoteit, H., 2023. Spatial-temporal prediction of minerals dissolution and precipitation using deep learning techniques: an implication to Geological Carbon Sequestration. *Fuel* 341, 127677. <https://doi.org/10.1016/j.fuel.2023.127677>.
- Tester, J.W., Worley, W.G., Robinson, B.A., Grigsby, C.O., Feerer, J.L., 1994. Correlating quartz dissolution kinetics in pure water from 25 to 625 °C. *Geochim. Cosmochim. Acta* 58, 2407–2420. [https://doi.org/10.1016/0016-7037\(94\)90020-5](https://doi.org/10.1016/0016-7037(94)90020-5).
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I., 2017. Attention is All You Need. <https://doi.org/10.48550/ARXIV.1706.03762>.

- Vilarrasa, V., 2014. Impact of CO₂ injection through horizontal and vertical wells on the caprock mechanical stability. *Int. J. Rock Mech. Min. Sci.* 66, 151–159. <https://doi.org/10.1016/j.ijrmms.2014.01.001>.
- Wang, Y., Lin, G., 2020. Efficient deep learning techniques for multiphase flow simulation in heterogeneous porous media. *J. Comput. Phys.* 401, 108968. <https://doi.org/10.1016/j.jcp.2019.108968>.
- Wen, G., Li, Z., Azizzadenesheli, K., Anandkumar, A., Benson, S.M., 2022. U-FNO—An enhanced Fourier neural operator-based deep-learning model for multiphase flow. *Adv. Water Resour.* 163, 104180. <https://doi.org/10.1016/j.advwatres.2022.104180>.
- Wu, H., Jayne, R.S., Bodnar, R.J., Pollyea, R.M., 2021. Simulation of CO₂ mineral trapping and permeability alteration in fractured basalt: Implications for geologic carbon sequestration in mafic reservoirs. *Int. J. Greenh. Gas Control.* 109, 103383. <https://doi.org/10.1016/j.ijggc.2021.103383>.
- Xie, X., Wang, X., Zhao, P., Hao, Y., Xie, R., Liu, H., 2023. Learning time-aware multiphase flow fields in coal-supercritical water fluidized bed reactor with deep learning. *Energy* 263, 125907. <https://doi.org/10.1016/j.energy.2022.125907>.
- Xin, Y., Yoo, H., Min, K.-B., Rutqvist, J., 2024. Coupled geomechanical analysis of irreversible compaction impact on CO₂ storage in a depleted reservoir. *Int. J. Greenh. Gas Control.* 135, 104158. <https://doi.org/10.1016/j.ijggc.2024.104158>.
- Xiong, W., Wells, R.K., Menefee, A.H., Skemer, P., Ellis, B.R., Giammar, D.E., 2017. CO₂ mineral trapping in fractured basalt. *Int. J. Greenh. Gas Control.* 66, 204–217. <https://doi.org/10.1016/j.ijggc.2017.10.003>.
- Xu, T., Sonnenthal, E., Spycher, N., Pruess, K., 2008. TOUGHREACT User's Guide: A Simulation Program for Non-isothermal Multiphase Reactive Geochemical Transport in Variably Saturated Geologic Media, V1.2.1 (No. LBNL-55460-2008, 943451). <https://doi.org/10.2172/943451>.
- Xu, W., Chen, R., Hu, Y., Chen, Y., Li, J., Zhuang, D., 2024. Effects of roughness on density-driven convection of dissolved CO₂ in discrete fracture-matrix system. *Comput. Geotech.* 166, 105996. <https://doi.org/10.1016/j.compgeo.2023.105996>.
- Yan, X., Lin, J., Wang, S., Zhang, Z., Liu, P., Sun, S., Yao, J., Zhang, K., 2024. Physics-informed neural network simulation of two-phase flow in heterogeneous and fractured porous media. *Adv. Water Resour.* 189, 104731. <https://doi.org/10.1016/j.advwatres.2024.104731>.
- Yang, Y., Dunham, E.M., 2021. Effect of Porosity and Permeability Evolution on Injection-Induced Aseismic Slip. *J. Geophys. Res. Solid Earth* 126, e2020JB021258. <https://doi.org/10.1029/2020JB021258>.
- Yeh, G., Tripathi, V.S., 1991. A model for simulating transport of reactive multispecies components: model development and demonstration. *Water Resour. Res.* 27, 3075–3094. <https://doi.org/10.1029/91WR02028>.
- Zakharova, N.V., Goldberg, D.S., Sullivan, E.C., Herron, M.M., Grau, J.A., 2012. Petrophysical and geochemical properties of Columbia River flood basalt: Implications for carbon sequestration. *Geochem. Geophys. Geosyst.* 13, 2012GC004305. <https://doi.org/10.1029/2012GC004305>.